MIL-A-55503(EL)
9 JUNE 1966

MILITARY SPECIFICATION

ATTITUDE HEADING REFERENCE SET AN/ASN-76

1. SCOPE

1.1 This specification covers the requirements of one type of Attitude Heading Reference Set, designated AN/ASN-76.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

FEDERAL

QQ-P-416	Plating, Cadmium (Electrodeposited)

MILITARY

MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-C-5541	Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-C-6781	Control Panel: Aircraft Equipment, Rack or Console Mounted
MIL-S-6872	Soldering Process, General Specification For
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for

FSC-6605

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MIL-P-7788	Plate, Plastic, Lighting
MIL-M-7793	Meter, Time Totalizing
MIL-A-8625	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-M-13231	Marking of Electronic Items
MIL-S-20708	Synchros, 60 and 400 Cycle, General Specification
MIL-F-25173	Fastener, Control Panel, Aircraft Equipment
MIL-D-25521	Detector, Magnetic, Azimuth DT-309/AJH
MIL-C-26500	Connectors, General Purpose, Electrical Miniature, Circular Environment Resisting, 200°C Ambient Temperature
MIL-R-26667	Reliability and Longevity Requirements, Electronic Equipment, General Specification for

STANDARDS

MIL-STD-143	Specifications and Standards Order of Precedence for the Selection of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-721	Definition of Terms for Reliability Engineering
MIL-STD-781	Reliability Test, Exponential Distribution
MIL-STD-831	Preparation of Test Reports
MS 17322	Meter, Time Totalizing Miniature, Digital 115 Volt 400 Cycle
MS 24264	Connectors, Receptacle, Electric Flange Mount, Miniature
MS 24266	Connectors, Plug, Electrical - Straight
MS 27464	Mounting Base for ASK-26/A24C-27 Electronic Control Amplifier
MS 33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS 33586	Metals, Definition of Dissimilar

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(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Components.- The Attitude Heading Reference Set AN/ASN-76 shall consist of the following:

a. Gyroscope, Displacement, CM-1151/ASN-76

b. Amplifier, Electronic Control AM-4682/ASN-76

c. Controller, Reference Set C-8099/ASN-76

3.2 Selection of specifications and standards.- Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials.-

3.3.1 Fungus-proof materials.- Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. If used within a hermetically sealed enclosure, fungicidal treatment shall not be required.

3.3.2 Metals.- Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to conditions likely to be encountered in storage or normal service.

3.3.2.1 Dissimilar metals.- Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS 33586.

3.3.3 Nonmagnetic materials.- Nonmagnetic materials shall be used for all parts of the equipment except where magnetic materials are essential.

3.3.4 Corrosive fumes.- Materials, as installed and used in the equipment and under the service conditions specified herein, shall not liberate deleterious fumes.

3.3.5 Protective treatment.- When materials are used in the construction of the equipment that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they

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shall be protected against such deterioration in a manner that will in no way prevent compliance with the requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4 Design and construction.- The equipment shall be designed and constructed to meet the requirements specified herein and to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. Corresponding components of all systems procured shall be interchangeable between systems. No gimbal binding under differential pressure conditions shall occur should a leak develop in service.

3.4.1 The components shall be designed to accept and provide input and output signals as specified herein.

3.4.2 Maintainability.- The following maintainability requirements shall be considered in the design of the control assembly.

a. Minimization of the complexity of maintenance tasks (for example: calibration, adjustments, inspection, etc.) by maximum use of simple design which includes optimum interchangeability and use of standardized equipment or commercially available items.

b. Rapid and positive recognition of equipment malfunction or marginal performance.

c. Rapid and positive identification of the replaceable defective part, assembly, or component.

d. Minimum numbers and types of tools and test equipment (special and standard) needed to perform maintenance.

e. Optimum accessibility in all systems, equipments, and components requiring maintenance, inspections, removal, or replacement.

3.4.3 Reliability. The contractual mean-time-between-failures (MTBF) for the Attitude Heading Reference Set and its components shall be as follows:

Attitude Heading Reference Set	1,120 hours
Displacement Gyroscope	2,240 hours
Electronic Control Amplifier	3,440 hours
Reference Set Controller	10,000 hours

The MTBF shall be based upon a discrimination ratio of 1.50 as defined in MIL-STD-781 and a confidence level of 90 percent. The MTBF shall apply when at least four systems or eight components are involved.

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3.4.3.1 Longevity.- The Attitude Heading Reference Set and its components shall have an operating life span (equipment longevity as defined in MIL-R-26667) of not less than 2,000 hours before they fail to meet the MTBF index. Those components that are subjected to reliability tests for 2,000 hours or longer shall not be subjected to a longevity test.

3.5 Displacement gyroscope.- The displacement gyroscope, including associated electronics, shall consist of a vertical and a directional gyro mounted in multiple gimbals as described in the following paragraphs. This gimbal arrangement is used in order to avoid the gimbal lock position.

3.5.1 Vertical gyro.- The vertical gyro shall have sufficient angular momentum and be so designed as to meet the performance specified herein when operated throughout the temperature, vibration, and acceleration ranges.

3.5.2 Directional gyro.- The directional gyro shall have sufficient angular momentum and be of such design as to provide the performance specified herein when operated throughout the temperature, vibration, and acceleration ranges.

3.5.3 Gimbal system.- The displacement gyroscope configuration shall consist of a two-axis directional gyro vertically stabilized by a two-axis vertical gyro.

3.5.3.1 Inner directional gyro gimbal.- The first (inner) directional gyro gimbal (gyro wheel housing) shall be stabilized by the angular momentum of the gyro. Its pivot axis shall be the first directional gyro axis of freedom, shall be perpendicular to the spin axis of the gyro, and normally shall lie in the horizontal plane. The slaving torquer shall act about this axis. The gimbal shall have a minimum of $\pm 30^{\circ}$ of freedom about its axis. Gimbal stops shall be located at $\pm 35^{\circ}$ $\pm 5^{\circ}$.

3.5.3.2 Directional gyro heading gimbal.- The second directional gyro gimbal shall be stabilized by the angular momentum of the gyro. Its pivot axis shall be the second directional gyro axis of freedom and shall be aligned with the vertical. The heading output synchro and the leveling torquer shall act about this axis. The gimbal shall be capable of continuous rotation about its axis.

3.5.3.3 Third gimbal.- The third gimbal shall be the pitch gimbal of the vertical gyro.

3.5.3.3.1 Pitch gimbal.- The first vertical gyro axis of freedom shall be stabilized by the angular momentum of the vertical gyro, shall be normally the pitch axis, and shall normally lie in a horizontal plane. The roll erection torquer shall act about this axis. The gimbal shall incorporate gimbal stops at $\pm 3^{\circ}$ $\pm 2^{\circ}$ for controlled precession during maneuvers through the vertical.

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A sensor for control of erection under linear acceleration shall be provided. (See para 3.6.4.2).

3.5.3.4 Fourth (outer) gimbal.- The fourth gimbal shall be the roll gimbal of the vertical gyro.

3.5.3.4.1 Roll gimbal.- The second vertical gyro axis of freedom shall be stabilized by the angular momentum of the vertical gyro, shall be the roll axis and shall be aligned with the fore-aft axis of the aircraft. The pitch erection torquer shall act about this axis. The gimbal shall have unlimited freedom throughout 360° of roll. A sensor for control of erection under turn accelerations shall be provided. (See para 3.6.4.1).

3.5.4 Erection.- The erection mechanism shall inoprporate separate systems for pitch and roll. Erection shall be accomplished by electrolytic switches and torque motors on the gimbals. The erection system shall provide both initial and final erection and shall erect the gyro from any attitude to the vertical as determined by the vertical sensing electrolytic switches.

3.5.4.1 Erection cycle.- The erection cycle shall consist of a 20-second initial direct erection followed by 100 seconds of fast precession erection, after which the system shall revert to the normal precession erection rate. Direct erection occurs when the pitch erection torquer is utilized to align the roll gimbal and the roll erection torquer is utilized to align the pitch gimbal. This can be accomplished when the angular momentum of the gyro is small (rotor is essentially at a standstill).

3.5.4.2 Initial erection.- Initial erection of the vertical gyro shall be such that it will assume a position within 1° of vertical in pitch and roll within 2 minutes or less after power is applied. The rotor shall be at a standstill and the spin axis may be in any possible attitude at the time power is applied. The vertical gyro shall align its spin axis with the direction of gravity within $\pm 1/4°$ in 2-1/2 minutes or less.

3.5.4.3 Erection rates.-

3.5.4.3.1 Normal erection rate.- The normal precession rate of erection about the roll axis shall be $1.25° \pm .5°$ per minute, and the pitch axis shall be $5° \pm 1°$ per minute, for deflections of spin axis greater than 0.2° from vertical. Actual rates shall be as shown on figure 1. Pitch erection sensing shall be maintained when displacing the gyro in roll $\pm 45°$ or less.

3.5.4.3.2 Fast erection.- Means shall be provided to accomplish faster erection of the vertical gyro to the apparent vertical of both axis of the gyro simultaneously. Fast erection excitation shall be made to terminal 5 and 6 of

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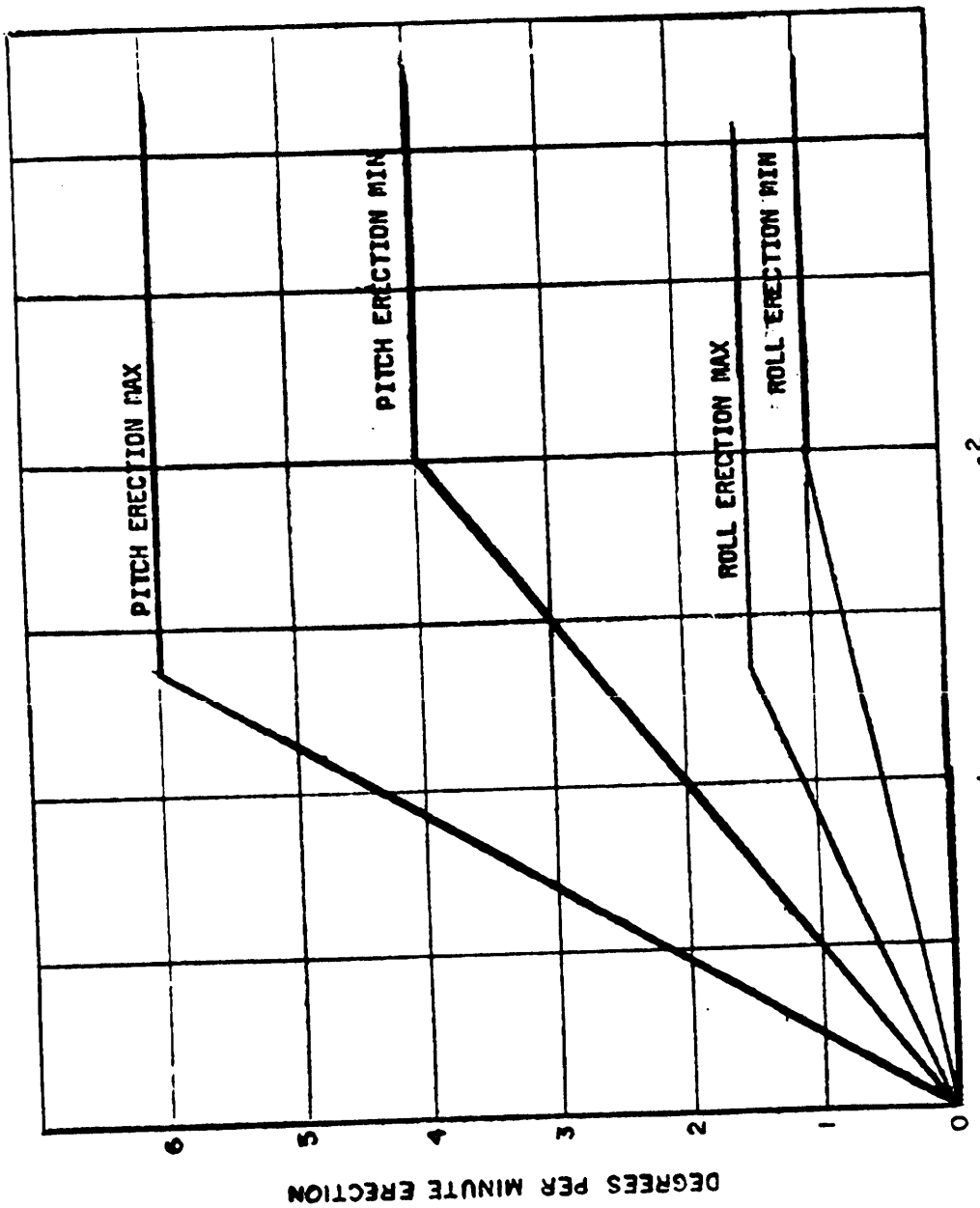


Figure 1. Erection Rate as a Function of Deflection

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Table I. The fast erection rate shall be not less than 12° per minute in either axis for a 78V ± 3V 400 cps signal to pins 5 and 6.

TABLE I. DISPLACEMENT GYROSCOPE CONNECTOR

MS 24264R16B-24PNX to mate with MS 24266R16B-24SN

TERMINAL	FUNCTION
1 (S1)	Heading output, ϕA excited
2 (S2)	Heading output, ϕA excited
3 (S3)	Heading output, ϕA excited
4	Erection warning
5	Pitch erection
6	Roll erection
7	Source of 26V 400 cps ϕC
8	Fast erection excitation
9	Turn acceleration sensor
10 (S1)	Pitch output, ϕA excited
11 (S2)	Pitch output, ϕA excited
12 (S3)	Pitch output, ϕA excited
13 (S1)	Roll output, ϕA excited
14 (S2)	Roll output, ϕA excited
15 (S3)	Roll output, ϕA excited
16	Linear acceleration sensor
17	N. Latitude control and slaving
18	S. Latitude control and slaving
19	Spare
20	Spare
21	Ground-neutral
22	115V ϕA
23	115V ϕB
24	115V ϕC

3.5.4.4 Roll and pitch erection cutout.- The design shall be such that when terminals 6 and 7 of Table I are connected externally, roll erection will be in effect and when terminals 5 and 7 are connected externally, pitch erection will be in effect. When all three pins are connected externally, both roll and pitch erection shall be functioning. The erection mechanism shall be so designed that when terminal 6 is disconnected externally, roll erection will be cut out and when pin 5 is disconnected externally, pitch erection will be cut out.

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3.5.4.5 Power interruption.- If a power interruption of up to 20 seconds occurs, the erection system shall revert to the fast precession erection rate, given in paragraph 3.5.4.3.2 for 120 seconds. The vertical gyro will align its spin axis with the direction of gravity within ±1° in 2½ minutes or less. If power is interrupted for more than 40 seconds, the displacement gyroscope will require 20 minutes without power before it can be guaranteed to meet the initial erection requirements specified in paragraph 3.5.4.2.

3.5.5 Vertical gyro synchro outputs.- One transmitting synchro shall be mounted on each vertical gyro axis of freedom. The synchros shall provide displacement signals of gimbal movement about their respective axes at a 1 to 1 ratio to the amplifier. The nominal electrical characteristics shall be in accordance with Table II. The synchros shall be internally excited from ϕA.

TABLE II. SYNCHRO CHARACTERISTICS

Input voltage	26V. ±1% 400 cps ±2%
Input current	80 ma ± 15%
Input power	0.44 w ±30%
Output voltage (max)	11.8V ±5%
Sensitivity	206 mv per deg ±5%
Phase shift	7.5° ±3°
D-C rotor resistance	43 ohms ±15%
D-C stator resistance	30 ohms ±15%
Impedance	Z_{ro}= 69 /j318 ohms ±20%
Impedance	Z_{so} = 29 /j60 ohms ±20%
Impedance	Z_{rss} =142 /j53 ohms ±20%
Maximum null	50 mv
Accuracy (max error spread)	14 minutes

3.5.5.1 Pitch synchro.- When terminal 11 of Table I is connected to terminal 21, and the gimbals positioned for zero pitch displacement, the R.M.S. voltage across terminals 10 and 12 shall be a minimum and the R.M.S. voltage from terminal 10 to terminal 22 shall be greater than the R.M.S. voltage from terminal 22 to terminal 21. When the gimbals are rotated in a nose-up direction, the R.M.S. voltage from terminal 10 to terminal 11 shall increase before decreasing.

3.5.5.2 Roll synchro.- When terminal 14 of Table I is connected to terminal 21, and the gimbals positioned for zero roll displacement, the R.M.S. voltage across terminals 13 and 15 shall be a minimum and the R.M.S. voltage from terminal 13 to terminal 22 shall be greater than the R.M.S. voltage from terminal 22 to terminal 21. When the gimbals are rotated in a right-wing-down direction, the R.M.S. voltage from terminal 13 to terminal 14 shall increase before decreasing.

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3.5.6 Vertical drift.- The vertical gyro assembly shall be so designed that the maximum free drift, noted after compensation for the earth's rotation, shall be not greater than 0.20° per minute at 25° ±10°C and 0.25° per minute over the temperature range specified herein.

3.5.7 Power-off freedom.- When power is removed for time durations of 20 seconds or less, the vertical gyro shall still have freedom about its first and second gimbal axes and shall not be displaced from the vertical during bank angles from 0 to ±80° or pitch angles from 0 to 80° during the power-off condition.

3.5.8 Directional gyro leveling.- The inner gimbal shall be slaved by means of the leveling torquer controlled by a suitable nonpendulous sensor so that the spin axis of the directional gyro is perpendicular to the heading axis. This leveling rate shall have a rate that will level the gyro within 1/4° with respect to vertical gyro platform within 50 seconds after power is applied.

3.5.9 Directional gyro slaving.- The directional gyro shall be precessed about its heading axis to an external heading signal by means of the slaving torquer. The normal slaving rate shall be 1° to 2° per minute.

3.5.10 Heading transmitting synchro.- One transmitting synchro shall be mounted on the heading axis of freedom. The synchro shall provide displacement signals of gimbal movement about the heading axis at a 1 to 1 ratio to the amplifier. The nominal electrical characteristics shall be in accordance with Table II.

3.5.10.1 Synchro rotation.- The gimbals shall be positioned to obtain a minimum voltage across terminals 1 and 3 and then rotated in a right-turn direction. The voltage from terminal 1 to terminal 2 shall increase before decreasing.

3.5.11 Balance.- The directional gyro shall not be balanced nor compensated internally for any particular lattitude other than the equator. Therefore, the apparent drift due to the earth's rotation is equal to the product of the sine of the latitude of operation and 15° per hour in a counterclockwise (ccw) direction in the Northern Hemisphere and in a clockwise (cw) direction in the Southern Hemisphere.

3.5.12 Maximum drift.- The maximum drift of the directional gyro shall not exceed ±1.0° per hour from the computed apparent drift for the latitude of operation at room temperature.

3.5.12.1 Random drift.- With normal distribution as defined in MIL-STD-721 the random drift of the directional gyro shall not exceed ±0.25° per hour - one sigma.

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3.5.13 Weight.- The weight of the displacement gyroscope shall not exceed 10.5 pounds including the vibration isolator. The vibration isolator is part of the displacement gyroscope.

3.5.14 Dimensions.- The dimensions of the displacement gyroscope shall be in accordance with figure 2.

3.5.15 Mounting.- The mounting over all dimensions, and center of gravity location of the displacement gyroscope shall be in accordance with figure 2 and shall be such that the displacement reference will not deflect by more than $\pm 0.1^{\circ}$ in pitch, roll, or heading as a result of accelerations or any vibration between 0 to 15 cps with a double amplitude of 0.020 inch. The torsional resonant period shall be greater than 15 cps in pitch, roll and yaw.

3.5.16 Case Alignment.- If the flat portion of the bottom of the vibration isolator is level, the pitch and roll synchro outputs shall be $0^{\circ} \pm 1^{\circ}$ under dynamic conditions. The dynamic condition is specified to eliminate the influence of static friction on the system. A suitable dynamic condition would be a 6 cycle per minute scorsby motion of $1\frac{1}{2}^{\circ}$ in pitch, roll and yaw.

3.5.17 Drift adjustment.- Provisions shall be made to adjust the displacement gyroscope heading drift rate. This adjustment shall be capable of changing the drift rate a minimum of $\pm .8^{\circ}$ and a maximum of $\pm 2.8^{\circ}$ per hour.

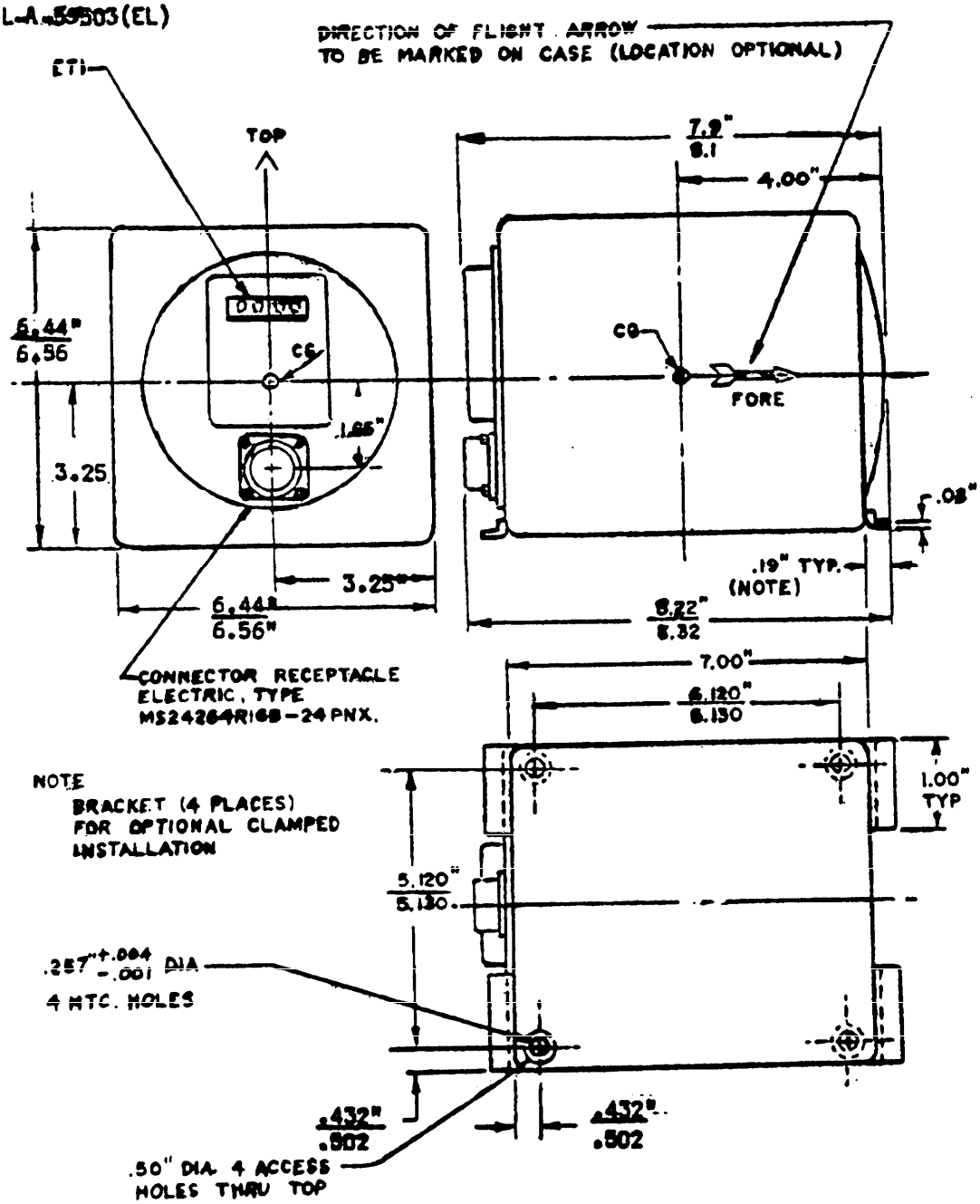
3.5.18 Elapsed time indicator.- The displacement gyroscope shall contain an elapsed time indicator conforming to MS17322 and MIL-M-7793 having a range of 9,999 hours. The ETI shall be located as shown on figure 2 and shall be visible without disassembly of the gyroscope case.

3.6 AM-4682/ASW-76 Electronic control amplifier.- The electronic control amplifier shall contain the heading servo amplifier, slaving amplifier, deviation compensator, turn and acceleration control features, fast synchronization circuitry, malfunction monitoring circuitry, and associated power supplies. It also shall function as a system interconnection box with connectors as shown in Tables III, IV, V, and VI.

3.6.1 Amplifier channels.- The amplifier channels shall be modular plug-in type and interchangeable, where practicable. The amplifier shall not incorporate electronic tubes. Where practical, integrated parallel circuits shall be used so that partial failure will not result in total loss of operation.

3.6.2 Fast synchronization and heading set drive.- An electro-mechanical mechanism shall be provided to positon the heading differential control transformer. The mechanism shall cause synchronization of the heading servo channel to the azimuth detector when operating in the slaved mode and shall position the heading servo to any desired reference when the heading system is operating in the DG mode.

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UNLESS OTHERWISE SHOWN DECIMAL DIMENSIONS ± .03

Figure 2. Displacement Gyroscope CN-1151/ASN-76

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The mechanism shall have a drive rate continuously variable in either direction up to 30° per second as selected by the heading set control specified in paragraph 3.7.2. The mechanism shall be suitably damped to permit accurate synchronization and slection of the desired reference.

3.6.2.1 Push-to-synchronization circuit.- The design of the circuit shall be such that when the heading set control (See para 3.7.2.1) is pressed and held and the heading system is in the slaved mode of operation, the system will synchronize to the compass heading within 6 seconds for all conditions except during the 2.5 min. immediately after power application due to erection cycle of vertical gyroscope.

3.6.2.2 Automatic autopilot decoupling.- The continuity between terminals 14 and 15 of Table IV shall be transferred to between 13 and 15 incident to and during synchronization, slewing, and compass mode conditions only. The design shall be such that the transfer must be made before the fast synchronizations can be actuated.

3.6.3 Servo.- The servo channel shall contain a servo system with a follow-up rate of 300° per second minimum to maintain synchronization with the displacement gyroscope signal throughout all maneuvers. The operation shall be smooth with no lag in excess of 0.1° per rpm up to 10 rpm and no oscillation, hunting, or jumping of a magnitude or frequency that will affect the operation of a high-performance autopilot flight control system shall occur. The undamped natural frequency shall be greater than 15 cps. The servo shall position the rotors of four data transmitter synchros.

3.6.4 Slaving and erection control.- Provisions shall be made for erection control and for deslaving the directional gyro. Erection control shall operate under all modes of operation. Slaving cut-out shall be effective when the equipment is in the slaved mode. Actuation circuitry for turn and fore-aft acceleration control shall be contained in the electronic control amplifier.

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TABLE III. ELECTRONIC CONTROL AMPLIFIER TO DISPLACEMENT GYROSCOPE

MS24264R16B-24S6X to mate with MS24266R16B-24P6

Terminal	Function
1(S1)	Heading input
2(S2)	Heading input
3(S3)	Heading input
4	Erection warning,
5	Linear acceleration signal
6	Turn acceleration signal
7	Pitch control
8	Roll control
9	Ground (neutral)
10	115V ØA
11	115V ØB
12	115V ØC
13	Fast erection excitation, 78V ØC
14	Spare
15	Spare
16	Spare
17	N. latitude control and slaving
18	S. latitude control and slaving
19(S1)	Pitch input
20(S2)	Pitch input
21(S3)	Pitch input
22(S1)	Roll input
23(S2)	Roll input
24(S3)	Roll input

3.6.4.1 Turn Acceleration cutoff.- An internal sensor on the displacement gyroscope and the associated circuitry shall be provided to permit cutoff of compass slaving during turns. Under standard conditions, cutoff shall occur at an aircraft lateral threshold acceleration of .065±.01g. The roll erection rate shall be 1-1/4° ±10°/min. and shall not be effected by the action of the sensor.

3.6.4.2 Linear acceleration control.- An internal sensor on the displacement gyroscope and the associated circuitry shall be provided to permit control of pitch erection and compass slaving during horizontal fore-aft acceleration.

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Control shall occur at a aircraft longitudinal threshold acceleration of 0.065g ±.010. During longitudinal accelerations in excess of 0.065g, the pitch erection shall be reduced to 25 percent of the normal value and compass slaving shall be cut off.

TABLE IV.

ELECTRONIC CONTROL AMPLIFIER to REFERENCE SET CONTROLLER and AZIMUTH DETECTOR CONNECTOR

MS24264R16B-24SNX to mate with MS24266R16B-24PN

TERMINAL	FUNCTION
1	Heading set relay
2	Heading set input
3	26V ϕA
4	ϕ27V
5	Mode select (slaved)
6	Sync indicator
7	Sync indicator
8	Latitude correction - South
9	Latitude correction - North
10	Slaving common (ground)
11	Malfunction indicator excitation
12	Fast erection relay
13	Automatic pilot cutoff normally open
14	Automatic pilot cutoff normally closed
15	Automatic pilot cutoff common
16 (S1)	Heading detector input
17 (S2)	Heading detector input
18 (S3)	Heading detector input
19	Shield ground
20 (R1)	Heading detector excitation
21 (R2)	Heading detector ground
22	Spare
23	Controller ground
24	Mode selector (Comp)

3.6.5 Data transmitters.- Four synchro transmitters whose rotors are

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positioned by the heading servo shall be provided to transmit information. Each synchro transmitter shall have the electrical characteristics as shown in Table XII. Three of these transmitters shall be designated primary and identified as No's 1, 2 and 3. The fourth shall be designated secondary. Each transmitter will be capable of handling four high impedance loads, (Control Transformer), or one low impedance load (Synchro Receiver). The combine impedance of the loads shall be equal to or greater than the Z_{so} of Table VII.

3.6.5.1 Data transmitter zero and sensing.- Input transmitters for simulating heading, pitch, and roll signals from the displacement gyroscope shall be indexed and coded for electrical zero and rotation in accordance with MIL-S-20708. These transmitters shall be excited from 400 cps, phase A, with R1 high and R2 low. Stator connections to the amplifier shall be in accordance with Table III.

3.6.5.1.1 Heading transmitter.- Rotor and stator leads of each heading output transmitter are identified in Table V in accordance with the lead coding standards of MIL-S-20708. With R2 of each output transmitter connected to S2, and with R1 to R2 excited from 400 cps, any phase, turn the heading input transmitter to obtain a minimum voltage from S1 to S3 and a voltage greater than the excitation voltage from S1 to R1. Positive rotation of the input transmitter (to simulate a right turn) shall cause the stator voltage of each output transmitter to become maximum in the sequence S2-1, S1-3, S3-2.

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TABLE V. AM-4682/ASN-76 Electronic Control Amplifier
Power Input and Heading Output Connector

MS24264R18B-31 PMX to mate with MS24266R18B-31SN	
Terminal	Function
1 (S1)	Heading output No. 1
2 (S2)	Heading output No. 1
3 (S3)	Heading output No. 1
4 (S1)	Heading output No. 2
5 (S2)	Heading output No. 2
6 (S3)	Heading Output No. 2
7 (S1)	Heading output No. 3
8 (S2)	Heading output No. 3
9 (S3)	Heading output No. 3
10 (S1)	Heading output No. 4
11 (S2)	Heading output No. 4
12 (S3)	Heading output No. 4
13 (R1)	Heading excitation No. 1
14 (R2)	Heading excitation No. 1
15 (R1)	Heading excitation No. 2
16 (R2)	Heading excitation No. 2
17 (R1)	Heading excitation No. 3
18 (R2)	Heading excitation No. 3
19 (R1)	Heading excitation No. 4
20 (R2)	Heading excitation No. 4
21	Ext malfunction signal (pos.)
22	Ext malfunction signal (neg.)
23	Ground (neutral)
24	115V ØA
25	115V ØB
26	115V ØC
27	Malfunction control (common)
28	Spare
29	Spare
30	Malfunction control
31	Spare

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TABLE VI. AM-4682/ASN-76 Electronic Control Amplifier
Attitude Output Connector

MS24264R14B-15SMX to mate with MS24266R14B-15PM	
Terminal	**Function**
1 (S1)	Pitch output
2 (S2)	Pitch output
3 (S3)	Pitch output
4 (S1)	Roll output
5 (S2)	Roll output
6 (S3)	Roll output
7	Spare
8	Spare
9	Ground
10	115 ØA
11	115 ØB
12	115 ØC
13	External fast erection
14	Spare
15	Spare

TABLE VII. Synchro Transmitter Electrical Characteristics

Input voltage	26V 400 cps ± 1%
Input current	100 ma ± 15%
Input power	.54 w ± 30%
Output voltage (max)	11.8V ± 5%
Sensitivity	206 mv deg ± 5%
Phase shift	11° ± 3°.
D-C rotor resistance	700 ohms ± 15%
D-C stator resistance	10.4 ohms ± 15%
Impedance	Z_{ro} = 950 + j3,850 ohms ± 20%
Impedance	Z_{so} = 10 + j36 ohms ± 20%
Impedance	Z_{rss} = 1,550 + j420 ohms ± 20%
Max null voltage	75 mv
Accuracy (max error spread)	14 minutes

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3.6.5.1.2 Pitch and roll transmitters. Stator leads of the pitch and roll output transmitters are identified in table VI in accordance with the lead coding standards of MIL-S-20708. Position the pitch and roll input transmitters to electrical zero. The voltage from S1 to S3 shall become a minimum. Positive rotation of either input transmitter from electrical zero (to stimulate nose-up or right-wing-down) shall cause the stator voltages of the corresponding output transmitter to become maximum in the sequence S2-1, S1-3, S3-2.

3.6.6 Malfunctioning monitoring. Provisions shall be made for detecting the following faults:

(a) Supply voltage high (indication initiated at between +10 and +20 percent).

(b) Supply voltage low (indication initiated at between -10 and -20 percent).

(c) Loss of synchro excitation caused by an open in the power source or in the rotor of the gyroscope heading synchro transmitter.

(d) Loss of synchro signal transmission caused by an open in any two stator connections from the gyroscope synchro heading transmitter. This includes opens in the aircraft interconnecting wiring between the gyroscope and the amplifier.

(e) Heading servo follow-up displaced from null. (Indication initiated at between 2° and 3° error signal).

3.6.6.1 Malfunction indication.

3.6.6.1.1 A positive 28V d-c signal shall be removed from pin 21 of 2J1 and a negative 28V d-c signal pin 22 of 2J1 when any fault is signaled as specified in 3.6.6.

3.6.6.1.2 A set of normally open contacts shall be provided between pins 27 and 30 of 2J1. These shall close when any fault is signaled as specified in 3.6.6.

3.6.6.1.3 Combined malfunction indication. A 26V ±2V phase A signal shall be applied to pin 11 of 2J3 when any fault is signaled as specified in 3.6.6.

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3.6.6.2 Fast erection indication. The malfunction signals shall be present until the completion of the erection cycle and shall occur during manual fast erection, except during COMP mode operation.

3.6.6.3 28V d-c malfunction signals. The 28V d-c signals provided at pins 21 and 22 of 2J1 shall have a load capability of 150 milliamps.

3.6.7 Deviation compensation. A 24-point electrical heading deviation compensator shall be provided having a nominal adjustment range of $\pm 4^{\circ}$ and a repeatable accuracy of ± 9 minutes at each 15° compensation point. The deviation compensation assembly shall be modular and plug-in, allowing transfer to another electronic control amplifier without major disassembly of either amplifer and without loss of the stored compensation information.

3.6.8 Dimensions. The amplifier dimensions shall be in accordance with figure 3.

3.6.9 Weight. The weight of the amplifier shall not exceed 6.5 pounds.

3.6.10 Elapsed time indicator. The amplifier shall contain an elapsed time indicator conforming to MS17322 and MIL-M-7793 with a range of 9,999 hours. It shall be visible without any disassembly of the amplifier.

3.6.11 Mounting base. When required (see 6.2), a mounting base conforming to MS27464 shall be provided for the amplifier.

3.7 Reference set controller. A reference set controller, designed in accordance with MIL-C-6781, MIL-P-7788, except the paragraph entitled "Connector Location" shall not apply, and MIL-F-25173 shall provide the controls and indications necessary to operate the heading system. The reference set controller shall conform to figure 4. This unit may be gasket sealed. Panel lighting shall utilize a 5V a-c/d-c, type IV or V red lighting system on pin 21 of table VIII.

3.7.1 Function selector switch. A mode selector switch shall be provided for selecting any of the following modes of operation:

a. Slaved - In the slaved mode, the system shall operate as a gyro stabilized magnetic compass.

b. DG - In the DG mode, the set shall provide heading when operated independent of the azimuth detector as a latitude corrected directional gyroscope.

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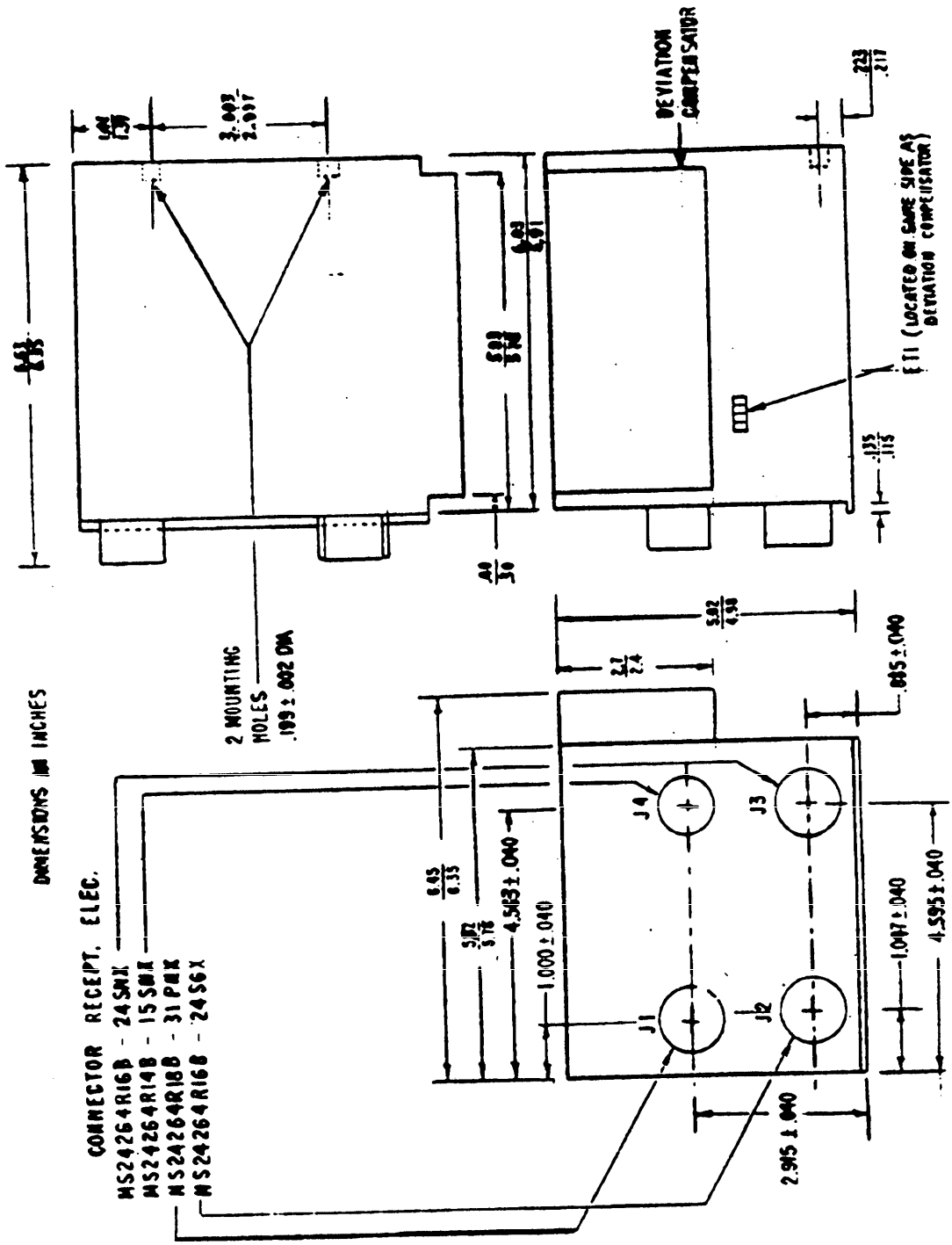


Figure 3. AM4682/ASN-76 Electronic Control Amplifier

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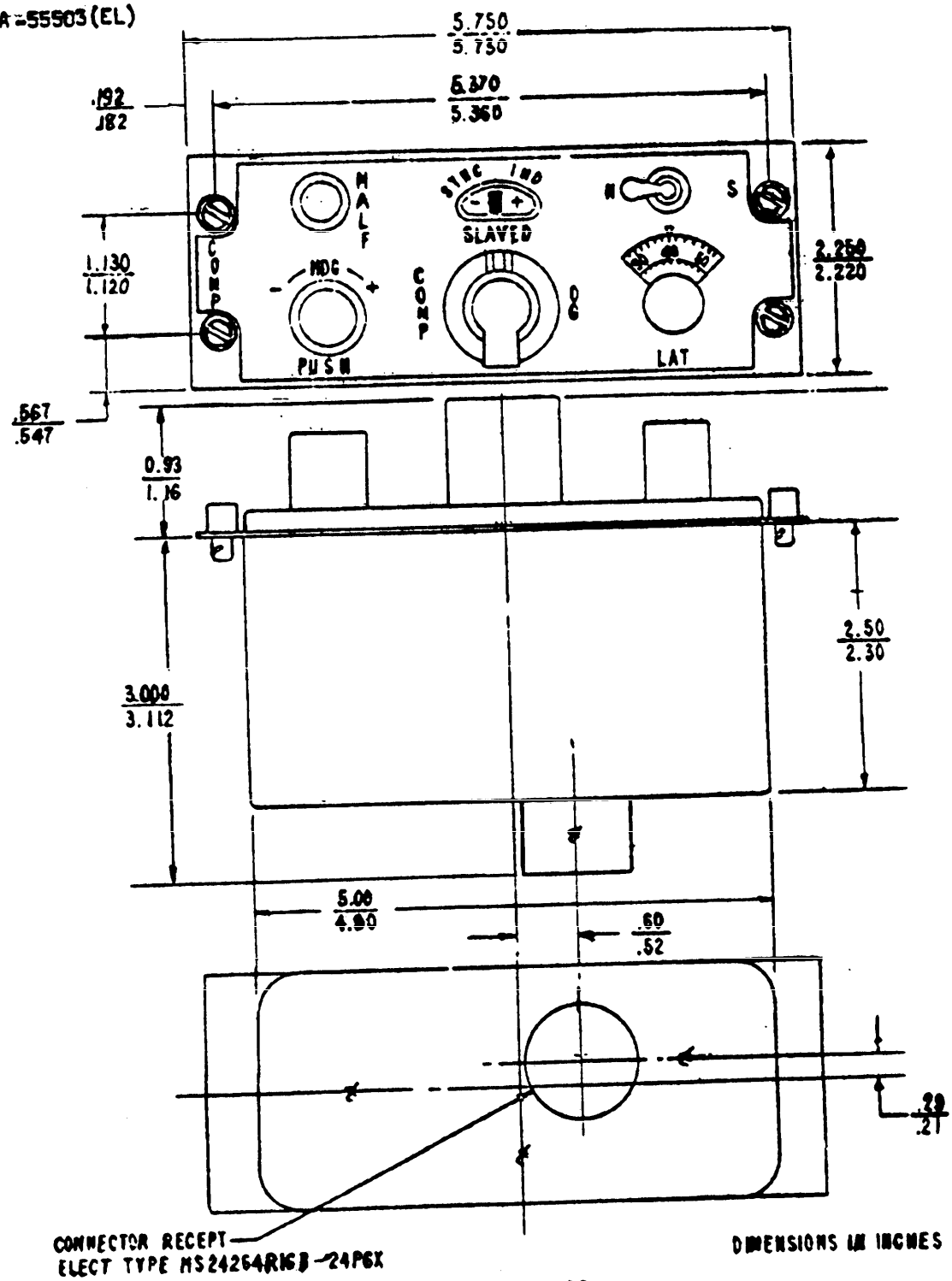


Figure 4. Reference Set Controller

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c. Comp (compass) - In the comp mode, the system shall be operated as an undamped magnetic compass slaved to the magnetic azimuth detector. During this mode, the continuity from pins 14 to 15 shall be terminated and replaced by the continuity from pins 13 to 15. This feature may be utilized in conjunction with an autopilot.

3.7.2 Synchronizer and heading set. A manual control shall be provided to effect the fast synchronization and heading set as specified in 3.6.2. When in the DG mode, the heading set control markings shall indicate in which direction the control is to be turned to provide a decreasing or increasing heading. The control shall be of the push-to-turn, spring return-to-center type.

3.7.2.1 Push-to-sync. When the heading set control is pushed, the PUSH-TO-SYNC circuit specified in 3.6.21 shall be actuated. It shall also operate the fast erection relay (pin 12 on 2J3).

3.7.3 Latitude correction controller. A manually set controller shall be provided to correct the system for the apparent drift of the gyro due to the earth's rotation. The selection of direction of the correction for the Northern or Southern Hemisphere shall be made by a hemisphere switch. The latitude dial shall be marked with latitude from 0° to 90° graduated in 2° increments, with major divisions and numerals every 10°. The correction shall be applied in both DG and slaved modes of operation.

3.7.4 Synchronizing indicator. A synchronizing indicator, other than signal lights, shall be provided to indicate synchronization to within ±1/4° between the compass system heading output and the magnetic detector, or electrical equivalent, when in the slaved mode of operation. When the system is not synchronized, the synchronizing indicator shall indicate in which direction the system must be corrected. The synchronizing indicator lighting and markings shall afford ease of viewing under all ambient lighting conditions.

3.7.5 Malfunction indicator. A push-to-test amber warning light shall be provided on the controller to serve as an indication of malfunctions as specified in 3.6.6.

3.7.6 Markings and dimensions

3.7.6.1 Marking. All markings shall be so durable as to withstand usage encountered in service. Unless otherwise specified, the style and proportions of numerals and letters shall conform to MS33558.

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3.7.6.2 Numerals. Each numeral shall be so distinctly marked that no doubt will exist as to which graduation the numeral applies.

3.7.6.3 Marking dimensions. The dimensions of the markings shall be as follows:

	Height or Length Inch ±0.005	Width of Line or Graduation Inch ±0.005
Letters or words	0.14	0.020
Numerals	0.12	0.020
2° graduations	0.10	0.015
10° graduations	0.14	0.020
"+" and "-"	0.12	0.020

3.7.6.4 Color. The background of the dial and all other markings not otherwise specified shall be finished in durable dull black.

3.7.7 Weight. The weight of the reference set controller shall not exceed 1.5 pounds.

3.8 Performance. The systems and each component shall be capable of meeting the test specified in section 4 under the following conditions:

a. Temperatures ranging from -54° to +71°C.

b. Extended storage at -65°C and 10 minutes at +98°C (except for the compass controller where the upper temperature limit is +85°C).

c. Dielectric strength of 375V ac or 500V dc, as specified.

d. Vibration according to 4.7.8 and 4.7.19.

e. Altitude pressure equivalent to 80,000 feet.

f. Shock forces of 15g.

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Table VIII. Reference Set Controller Connector to Electronic Control Amplifier

MS24264R16B-24P6X to mate with MS24266R16B-24S6	
Terminal	Function
1	Heading set switch
2	Heading set output
3	26V ØA
4	+27V
5	Mode select (slaved)
6	Sync indicator
7	Sync indicator
8	Latitude correction S
9	Latitude correction N
10	Slaving common
11	Malfunction indicator
12	Fast erection
13	Spare
14	Spare
15	Spare
16	Spare
17	Spare
18	Spare
19	Spare
20	Spare
21	Panel lights (red)
22	Panel lights (5V)
23	Ground
24	Mode select (comp)

g. Humidity from 0 to 100 percent.

h. Exposure to fungus enmyes.

i. Exposure to a simulated rainfall of 4 inches per hour.

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j. Salt spray for a period of 50 hours.

k. Exposure to blowing fine sand and dust particles.

l. Operation in explosive vapors.

m. Sealing thermal shock consisting of 8 cycles in water at 85° ±2°C and then in water at 5° ±2°C.

n. Immersion in water with a pressure of 2-1/2 inches Hg applied for 1 minute for hermetically sealed units.

o. Exposure to sudden temperature changes.

3.8.1 Radio noise interference. The attitude heading reference set and each component shall meet the radio noise interference requirements of MIL-I-6181.

3.9 Electronic components. The electronic components shall be in accordance with instructions provided by the procuring activity.

3.10 Electrical power requirements. The power required by the attitude heading reference set shall be 4-wire, 200/115V, 3-phase, 400-cps, wye (nominal) in accordance with MIL-STD-704 and as specified herein. The Reference Set Controller lighting shall utilize 5V ac or dc.

3.10.1 Electrical power variation. The equipment shall operate within specified performance limits from a category B power source as defined in MIL-STD-704.

3.10.2 Phase rotation. The phase rotation of the 400-cycle power required by the equipment shall be A-B-C with neutral as ground.

3.11 Electrical connectors. Unless otherwise specified, electrical connectors shall be in accordance with MIL-C-26500.

3.11.1 Test terminals. Adequate test terminals shall be provided on each component for external testing of the item.

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3.11.2 Bonding. Adequate bonding points shall be provided on each component of the system.

3.12 Cases. The cases shall be in accordance with the specified dimensions and shall provide a hermetically sealed enclosure when required for the internal mechanism. The design of the case shall be such that the internal mechanism may be removed from the case, replaced, and the case resealed without the use of special tools and fixtures unless they are approved by the procuring activity. The hermetic sealing shall be so accomplished that the seal will not be dependent upon materials that will be affected by the action of any atmosphere to which the equipment may be subjected and in such a manner that it will meet the test specified in section 4.

3.12.1 Hermetic sealing. The following hermetic sealing requirements shall apply except in those extreme cases where the filling medium or the pressure, or both, materially affects the performance of the instrument. Exceptions shall be based upon sound engineering facts and shall be presented to the procuring activity for approval.

3.12.1.1 Filling medium. The filling medium shall be a mixture of 98% helium and 2% nitrogen and shall contain a maximum of 0.006 milligram of water vapor per litre (dewpoint -65°C at filling pressure). The absolute pressure of the filling medium shall be approximately 1 or 1/2 atmosphere.

3.13 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-1000 shall govern the manufacturer's part number and changes thereto.

3.14 Soldering. Soldering shall be accomplished in accordance with MIL-S-6872.

3.15 Screw threads. Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

3.16 Weight. The weight of the equipment shall not exceed 18.5 pounds.

3.17 Finishes and protective coatings.

3.17.1 Aluminum alloy parts. Aluminum parts shall be protected with an anodic film conforming to MIL-A-8625, or a chemical film conforming to MIL-C-5541.

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3.17.1.1 Dials, small holes, and case inserts need not be anodized.

3.17.1.2 When abrasion resistance is a factor, chemical film in accordance with MIL-D-5541 shall not be used in lieu of anodizing.

3.17.2 Cadmium plating.- Cadmium plating shall be in accordance with QQ-P-416, and of a type and class to achieve the degree of protection required.

3.18 Identification of product.- Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-M-13231.

3.19 Dimensions and tolerances.- Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the system, they shall be held or limited accordingly.

3.20 Screw assemblies.- Assembly screws and bolts shall be tight. The word tight means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw or bolt or threads. The screws and bolts shall remain tight without adjustment throughout all vibration test in MIL-T-5422.

3.21 Riveting.- Riveting operations shall be performed carefully to insure that the rivets are tight and satisfactorily headed.

3.22 Gears.- Gear assemblies shall be properly aligned and meshed and shall operate without interference, tight spots, loose spots or other irregularities. Where required for accuracy adjustment, gear assemblies shall be free from back lash.

3.23 Cleaning.- The components shall be cleaned thoroughly of loose, spattered or excess solder, metal chips, or other foreign material after assembly. Burrs and sharp edges as well as resin flash which might crumble shall be removed.

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3.24 Compatibility. The set shall be compatible with the Magnetic Azimuth Detector, DT-309/AJN as specified in MIL-D-25521. Provisions shall be made for its excitation.

3.25 Test Reports

3.2.5.1 Content. Test reports showing compliance to the requirements of this specification are required and shall be in accordance with MIL-STD-831. These reports shall contain all test data showing compliance of each item to each test specified in paragraph 4, that is required for the item. The test data shall be numerical values where applicable instead of GO-NO-GO type of data. The test reports shall contain information, reasons, method of repair and the necessary retest of the equipment for each and every failure from the beginning of the individual test to the final test performed on the equipment. Reports should contain explanations and conclusions derived from consideration of the data when necessary.

3.2.5.2 Presentation. The test data shall be presented in tabular form in two ways.

(1) A list showing the numerical test data of all equipments for each test.

(2) A list showing the numerical test data of all tests for each equipment.

3.25.3 Delivery.

3.25.3.1 A partial list of test data and test reports shall be submitted to the procuring activity on a periodic basis which shall be specified by that activity.

3.25.3.2 A final list of all test data and a final test report shall be submitted to the procuring activity containing all the information on all the equipment on all of the test.

3.25.3.3 A reliability test report shall be submitted in accordance with MIL-STD-781.

3.26 Workmanship.- The equipment, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, brazing, painting, riveting, machine screw assemblies, and freedom of parts from burrs and sharp edges.

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4. QUALITY ASSURANCE PROVISIONS

NOTE: The word system refers to Attitude Heading Reference Set throughout this section.

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform or repeat any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of test.- The inspection and testing of the equipment shall be classified as follows:

a. Preproduction test

b. Acceptance test

4.3 Test conditions.-

4.3.1 Standard conditions.- Unless otherwise specified, the following standard test conditions shall be used:

Temperature	Room temperature +25° ± 10° C
Altitude	Atmospheric Pressure 725 ± 75mm of mercury (28.5± 3 in. of mercury)
Vibration	None
Humidity	Room ambient (From 0 to 90 percent relative humidity)
Attitude	Level within 1°
Input voltage	Three phase, 400 ±4 cps, wye, 115 ±1 V to common neutral, phase rotation A, B, C

4.3.2 Special test equipment.- The following special test equipment shall be used in testing.

4.3.2.1 Scorsby table.- A scorsby table, equivalent to IDEAL AEROSMITH Part Number 1412 shall make an excursion about each axis of ±1-1/2° at 5 to 7 cycles per minute with motion reversal once each minute.

4.3.2.2 Indexing fixture.- The indexing fixture shall be so designed that it can be indexed to ±90° in pitch and 360° in roll from a horizontal attitude and can be accurately positioned through 360° in heading.

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4.3.2.3 Servoed repeater.- The servoed repeater shall be scale graduated in 2-minute angular intervals or smaller and shall have a resolution of 1 minute or less.

4.3.2.3.1 Lead coding.- Connect the servoed repeater to excite the rotor and to receive the stator signals of a synchro transmitter that has been previously adjusted and coded for electrical zero and rotation per MIL-S-20708. Arrange these connections to obtain a zero angular readout with the transmitter on electrical zero, and an increasing angular readout as the transmitter stator voltages become maximum in the sequence S2-1, S1-3, S3-2. Code the servoed repeater leads R1, R2, S1, S2, S3 to correspond with the coding of the associated transmitter leads.

4.3.2.4 Simulated gyroscope transmitter.- Clifton Synchro Transmitter type CGH-11-B-2, or equivalent, mounted in an indexing head and adjusted and coded for electrical zero and rotation per MIL-S-20708. Rotor to be excited from 26 ±1V phase A with R1 "high" and R2 "low."

4.3.2.5 Simulated azimuth detector.- Clifton Synchro Transmitter type CGH-11-B-2, or equivalent, mounted in an indexing head and adjusted and coded for electrical zero and rotation per MIL-S-20708. Rotor to be excited from 230 ±10 mv 800 cps with R1 high and R2 low. The high side of the 800-cps excitation voltage is identified with the high side of the 400 cps phase A reference voltage as follows:

(a) The 800 cps excitation voltage passes through zero amplitude at the same instant that the 400 cps reference voltage passes through zero amplitude.

(b) The 800 cps excitation voltage increases negatively from zero amplitude at the same instant that the 400 cps reference voltage increases positively from zero amplitude.

4.3.2.6 Simulated compass controller.- Provides controls and signals simulating those functions of the controller required for conducting tests of the electronic control amplifier. An actual compass controller is acceptable.

4.3.2.7 Motor driven simulated gyroscope transmitter.- The motor driven simulated gyroscope transmitter shall be similar to the transmitter of 4.2.2.4 except that it shall be so designed that it can be driven at 5 rpm and 50 rpm ±10 percent.

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4.4 Preproduction Testing

4.4.1 Test Samples.- The test samples shall consist of the following: Three samples of each system (or component, as applicable) shall be subjected to the limited performance test (all the test specified in 4.7 except reliability and longevity) and the remaining samples shall be subjected to the reliability test only. The samples shall be identified with manufacturer's part number and such other information as required by the procuring activity.

EQUIPMENT	NO. OF SAMPLES
Attitude Heading Reference Set	9
Displacement Gyroscope	15
Electronic Control Amplifier	13
Reference Set Controller	15

4.4.2 Rejection and Retest

4.4.2.1 Limited Performance Test.- The equipment must meet all the requirements of the test as specified in 4.7 except reliability and longevity without failure. If any system (or component, as applicable) fails to meet any requirement, the equipment will be unacceptable. However if the equipment is repaired or replaced, this equipment may be retested. Any repaired or replaced equipment shall be subjected to all the limited performance test after repair or replacement, unless otherwise directed in writing by the contracting officer. In the repairing of the equipment, the equipment will not be redesigned or altered so that it is no longer identical to the other equipment under test. If this occurs all preproduction test will be unacceptable. If redesign is necessary or desirable, preproduction test on the newly designed equipment will be permissible. Consult Table IX and X for clarification and convenience.

4.4.2.2 Reliability Test.- Unless otherwise specified, the system or its components shall be subjected to the reliability test specified under 4.7.23. The total test time, number of failures permitted, and the reject criteria prior to truncation shall be determined in accordance with figures 5 through 8. The equipment being tested to the reliability test must be identical to the equipment being tested to the limited performance test.

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TABLE IX
SUMMARY OF TEST REQUIREMENTS

		Individual Test	Sampling Plan A	Sampling Plan B	Sampling Plan C	Sampling Plan D
Preproduction Test (9 to 15 Samples)	Limited Performance Test 3 of Prepro. Samples	X	X	X	X	X
	Reliability Test (Remainder of Prepro Sample) every 160 hrs.	X every 160 hrs.	X	X	X	X every 100 hrs.
Acceptance Test	Individual Test	X	X	X	X	X every 100 hrs.
	Sampling Plan A	X	X	X	X	X
	Sampling Plan B	X	X	X	X	X
	Sampling Plan C	X	X	X	X	X
	Sampling Plan D	X	X	X	X	X

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TABLE X
SUMMARY OF TESTS

REQUIREMENT	PARAGRAPH NR. OF REQUIREMENT	PARAGRAPH NR. OF TEST	CATEGORY OF TEST
Examination of Product	3.1 thru 3.26	4.7.1	Individual tests for all Components
Erection Cycle and Accuracy	3.5.4.2 and 3.5.4.5	4.7.2.1	Individual tests for D. Gyro
Heading Rotation	3.5.10	4.7.2.2	" "
Heading Drift Rate	3.5.12	4.7.2.3	" "
Vertical Drift Rate	3.5.6	4.7.2.4	" "
Erection Rate	3.5.43	4.7.2.5	" "
Heading Precession Rate	3.5.9 and 3.7.3	4.7.2.6	" "
Linear Acceleration Sensor Threshold	3.6.4.2	4.7.2.7	" "
Turn Acceleration Sensor Threshold	3.6.4.1	4.7.2.8	" "
Fast Erection Warning Signal	3.6.6.2	4.7.2.9	" "
Pitch Gimbal Freedom	3.5.3.3.1	4.7.2.10	" "
Mode Selection	3.7.1	4.7.3.1	Individual tests for Controller
Hemisphere Selector	3.7.3	4.7.3.2	" "
Latitude Control Current	3.7.3	4.7.3.3	" "
Synchronization Indicator Deflection	3.7.4	4.7.3.4	" "
Heading Set Operation	3.7.2	4.7.3.5	" "
Malfunction Indicator	3.7.5	4.7.3.6	" "
Panel Lighting	3.7	4.7.3.7	" "
Slaving Amp. Output and Gain	3.5.9 and 3.8	4.7.4.1	Individual tests for Amplifier

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TABLE X (Cont'd)

REQUIREMENT	PARAGRAPH NR. OF REQUIREMENT	PARAGRAPH NR. OF TEST	CATEGORY OF TEST	
Heading Compensation Range	3.6.7	4.7.4.2	Individual tests for Amplifier	
Heading Servo Scale Error, DC Mode	3.8	4.7.4.3	"	"
Heading Servo Scale Error, Slaved Mode	3.8	4.7.4.4	"	"
Heading Servo Following Rate	3.8	4.7.4.5	"	"
Heading Set Rate	3.6.2 and 3.8	4.7.4.6	"	"
Malfunction Monitoring Circuit Performance	3.6.6	4.7.4.7	"	"
Compass Mode Operation	3.7.1	4.7.4.8	"	"
Linear Acceleration Control Functions	3.6.4.2	4.7.4.9	"	"
Turn Acceration Cutoff Function	3.6.4.1	4.7.4.10	"	"
Heading Rotation Polarity	3.6.5.1.1	4.7.5.3.1	Sampling Plan A	
Roll Correspondence and Rotational Polarity	3.6.5.1.2	4.7.5.3.2	"	"
Pitch Correspondence and Rotational Polarity	3.6.5.1.2	4.7.5.3.3	"	"
Heading Set Function	3.7.4	4.7.5.3.4	"	"
Slaving Rate and Correspondence	3.5.9	4.7.5.3.5	"	"
System Heading Drift Rate	3.5.12	4.7.5.3.6	"	"
Torsional Response	3.5.15	4.7.2.11	Sampling Plan B	
Voltage and Frequency Variation	3.10	4.7.6	Sampling Plan B	
Power Consumption	3.8	4.7.7	"	"
Vibration Error	3.8d	4.7.8	"	"
Heading Drift Adjustment	3.5.17	4.7.24	"	"

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TABLE X (Cont'd)

REQUIREMENT	PARAGRAPH NR. OF REQUIREMENT	PARAGRAPH NR. OF TEST	CATEGORY OF TEST
Low Field Strength	3.8	4.7.9	Sampling Plan B
Slaved Dynamic Accuracy	3.8	4.7.10	" "
Radio Noise Interference	3.8.1	4.7.11	Sampling Plan C
Temperature-Altitude (Cycling)	3.8 a and e	4.7.12	" "
Humidity	3.8g	4.7.13	" "
Fungus	3.8h	4.7.14	" "
Rain	3.8i	4.7.15	" "
Salt Spray	3.8j	4.7.16	" "
Sand and Dust	3.8k	4.7.17	" "
Temperature Shock	3.8a	4.7.18	" "
Vibration Failure	3.8d	4.7.19	" "
Shock	3.8f	4.7.21	" "
Explosion	3.8L	4.7.20	" "
Crash Safety	3.8	4.7.22	" "
Reliability and Longevity	3.4.3	4.7.23	Sampling Plan D

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4.5 Acceptance tests.- Acceptance tests shall consist of the following:

(a) Individual tests.

(b) Sampling tests.

4.5.1 Individual tests.- Each Attitude Heading Reference Set or component, as applicable, shall be subjected to the following tests as described under 4.7.

(a) Displacement gyroscope.

(1) Examination of product.

(2) Erection cycle and accuracy.

(3) Heading rotation.

(4) Heading drift rate.

(5) Vertical drift rate.

(6) Erection rate.

(7) Heading precession rate.

(8) Linear acceleration sensor threshold.

(9) Turn acceleration sensor threshold.

(10) Fast erection warning signal.

(11) Pitch gimbal freedom.

(b) Reference Set Controller.

(1) Examination of product.

(2) Mode selection.

(3) Hemisphere selector.

(4) Latitude control current.

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(5) Synchronization indicator deflection.

(6) Heading set operation.

(7) Malfunction indicator.

(8) Panel lighting.

(c) Electronic Control Amplifier.

(1) Examination of product.

(2) Slaving amplifier output and gain.

(3) Heading compensation range.

(4) Heading servo scale error, DG mode.

(5) Heading servo scale error, compensated, slaved mode.

(6) Heading servo follow up rate.

(7) Heading set rate.

(8) Malfunction monitoring circuit performance.

(9) Compass mode operation.

(10) Linear acceleration control function.

(11) Turn acceleration cutoff function.

4.5.2 Sampling tests.-

4.5.2.1 Sampling plan A.- One system (or component) selected at random from each 25 or less produced on the contract or order shall be subjected to the following tests as described under 4.7. A simulated magnetic detector may be used.

(a) Individual tests.

(b) Heading rotational polarity.

(c) Roll correspondence and rotational polarity.

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(d) Pitch correspondence and rotational polarity.

(e) Heading set function.

(f) Slaving rate and correspondence.

(g) System heading drift rate.

4.5.2.2 Sampling plan B.- One component (or system) selected at random from each 100 or less produced on the contract or order shall be subjected to the following tests as described under 4.7.

(a) Sampling plan A tests.

(b) Voltage and frequency variation.

(c) Power consumption.

(d) Vibration error.

(e) Low field strength.

(f) Slaved dynamic accuracy.

(g) Heading Drift adjustment.

(h) Torsional response.

4.5.2.3 Sampling plan C.- Unless otherwise specified, 3 samples of each component shall be selected at random from the first 15 items on the contract or order and subjected to the following applicable tests as described under 4.7.

(a) Sampling plan B tests.

(b) Radio noise interference.

(c) Temperature-altitude (cycling).

(d) Humidity.

(e) Fungus.

(f) Rain.

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(g) Salt spray.

(h) Sand and dust.

(i) Temperature shock.

(j) Vibration failure.

(k) Shock.

(l) Explosion.

(m) Crash safety.

4.5.2.4 Rejection and retest.- When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined.

4.5.2.4.1 Individual tests may continue.- For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.5.2.4.2 Corrective action and retest.- Corrective action and retest shall be required for each failure that can be reasonably attributed to poor manufacture, contractor handling or design practices, or which adversely affects mission completion.

4.5.2.5 Sampling plan D.- Unless otherwise specified, the following number of units shall be selected and subjected to the reliability tests specified under 4.7.2.3, for the following indicated test periods. The test time for components is applicable when the components are tested separately.

EQUIPMENT	TEST TIME (HOURS)	NO. OF UNITS TESTED
Attitude Heading Reference Set	4,925	4 out of every 100
Displacement Gyroscope	9,850	8 out of every 100
Electronic Control Amplifier	15,100	8 out of every 100
Reference Set Controller	44,000	9 out of every 100

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4.5.2.5.1 Acceptance or rejection.- Accept or reject criteria shall be in accordance with figures 5 through 8 as applicable to the Attitude Heading Reference Set or components. Certified time records shall be maintained of the test time accumulated on the equipment after starting the environmental cycling specified in 4.7.2.3. In the event a continue-test decision applies after the completion of the test time specified in 4.7.2.5 the production quantity shall be considered as conditionally acceptable; however, the accumulated time and failures shall be added to data of the next production quantity for a composite accept-reject decision. If the continue-test condition persists, the data from successive production quantities shall be accumulated until the truncation point shown on figures 5 through 8 has occurred, at which time an accept-reject decision shall be made for the previous conditionally acceptable production quantity. The continue test region of figures 5 through 8 are inclusive of its own boundary line except the vertical line at the truncation point and this line is part of the accept region. Test time is the actual operating time of the equipment during the reliability test.

4.6 Defects in items already accepted.- The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall advise the procuring activity of all defects likely to be found and methods of correcting them.

4.7 Test methods.-

4.7.1 Examination of product.- The Attitude Heading Reference Set and components shall be examined to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.7.2 Displacement gyroscope.- The displacement gyroscope shall be tested as follows:

(a) Connect stator leads of heading, pitch, and roll servoed repeaters to the following connector pins:

HEADING	PITCH	ROLL
S1 to 1	S1 to 10	S1 to 13
S2 to 2	S2 to 11	S2 to 14
S3 to 3	S3 to 12	S3 to 15

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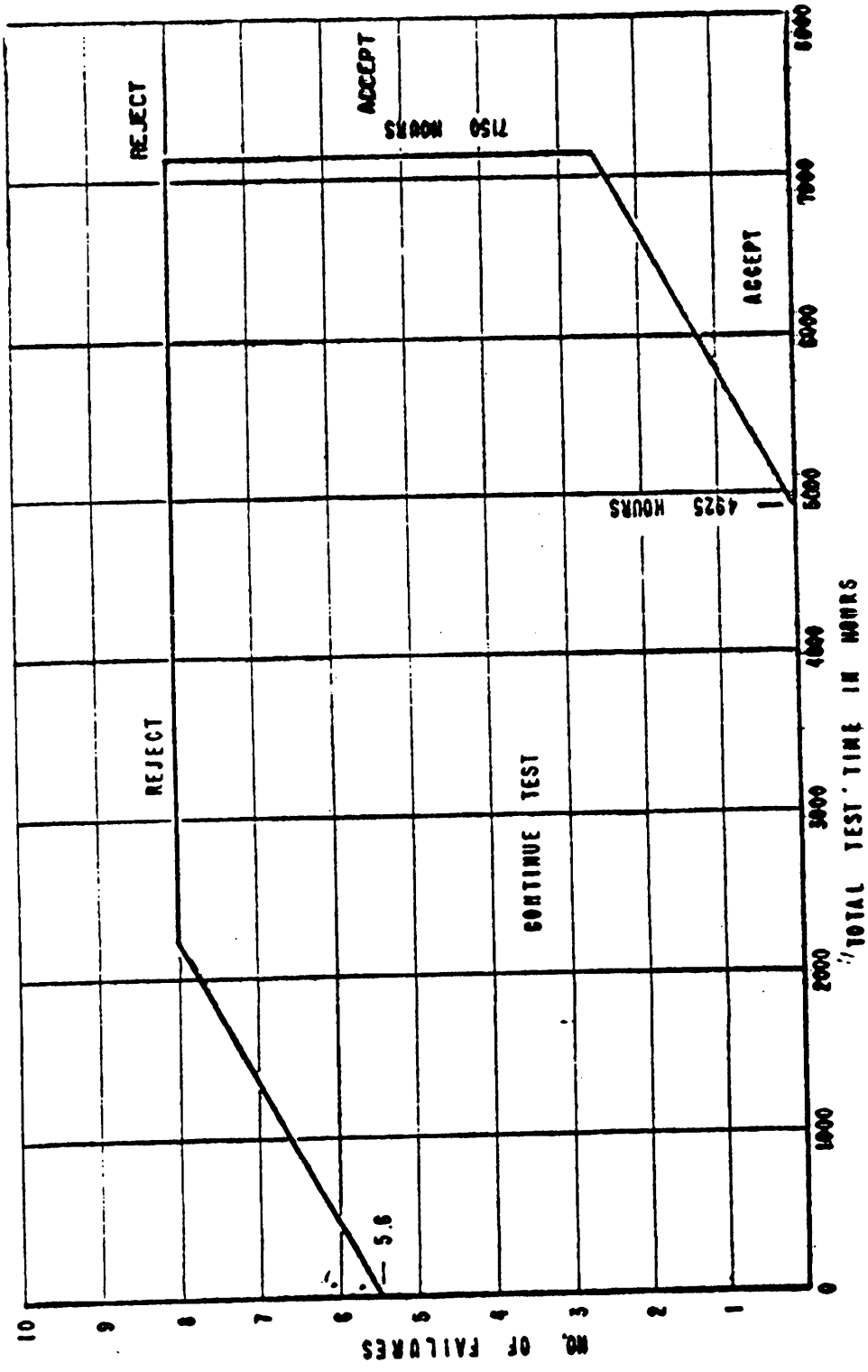


Figure 5. Set Accept-Continue-Test-Reject Criteria

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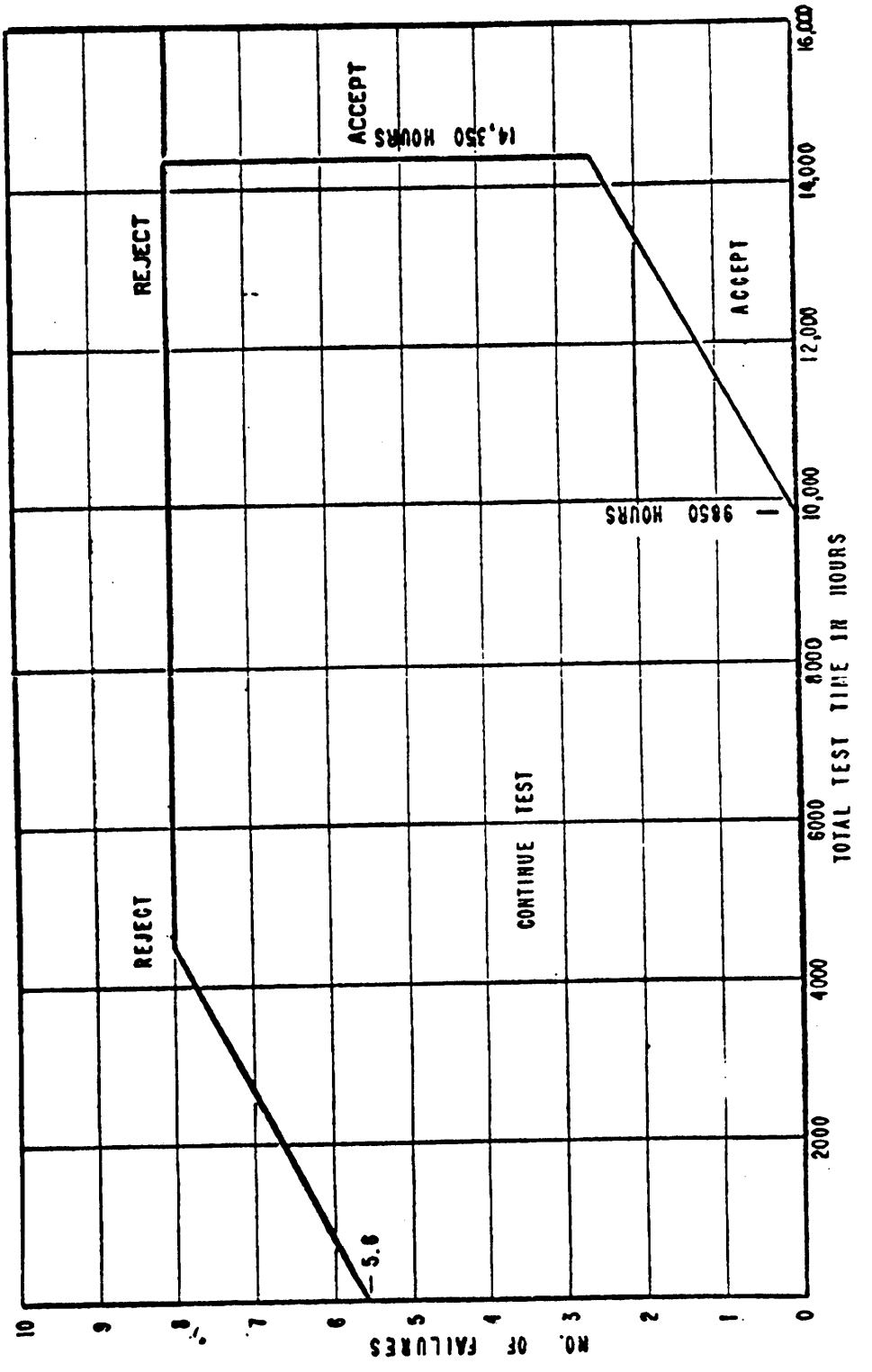


Figure 6. Gyroscope Reject-Continue-Test-Accept Criteria

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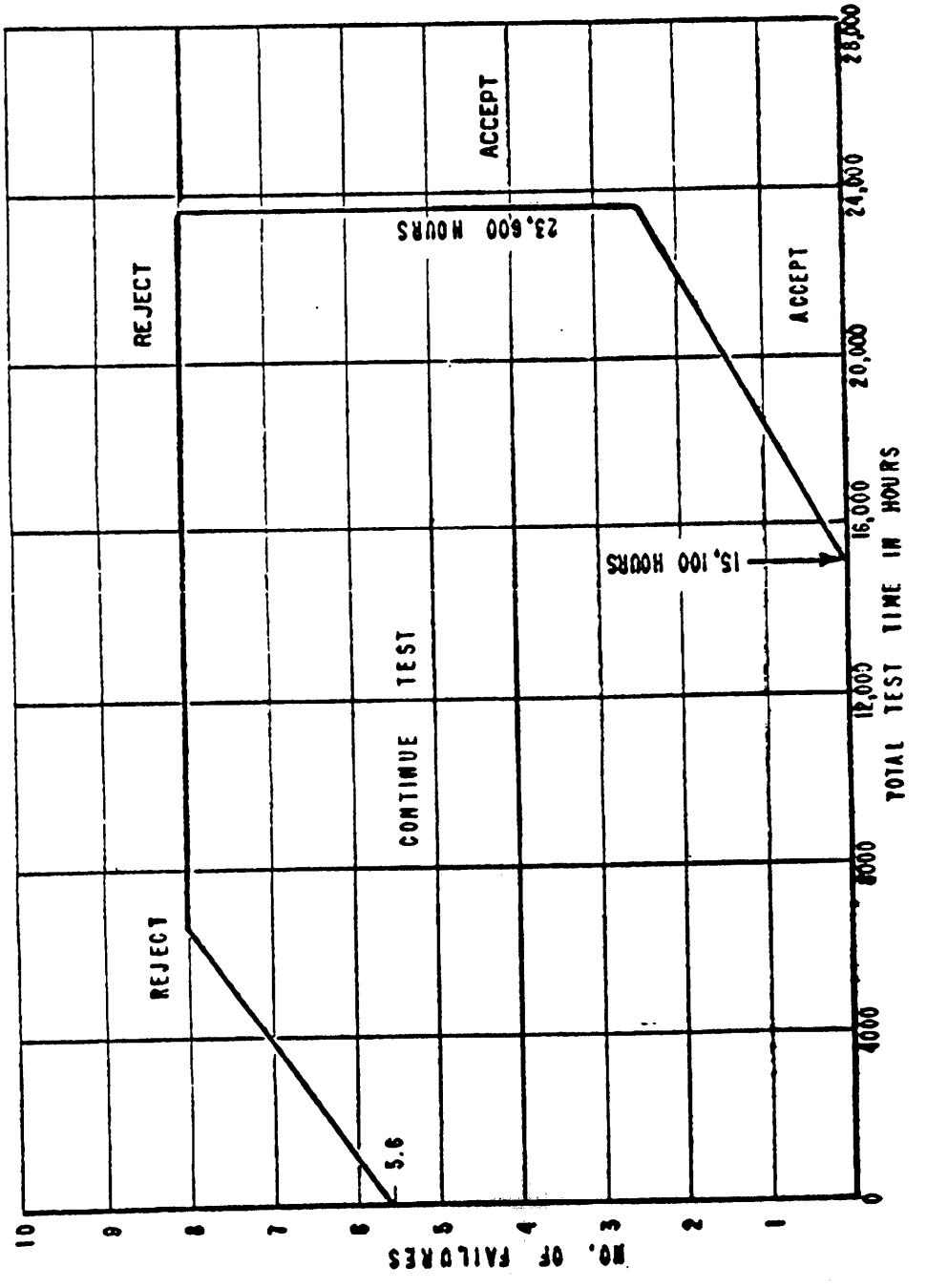


Figure 7. Amplifier Accept-Continue Test-Accept Criteria

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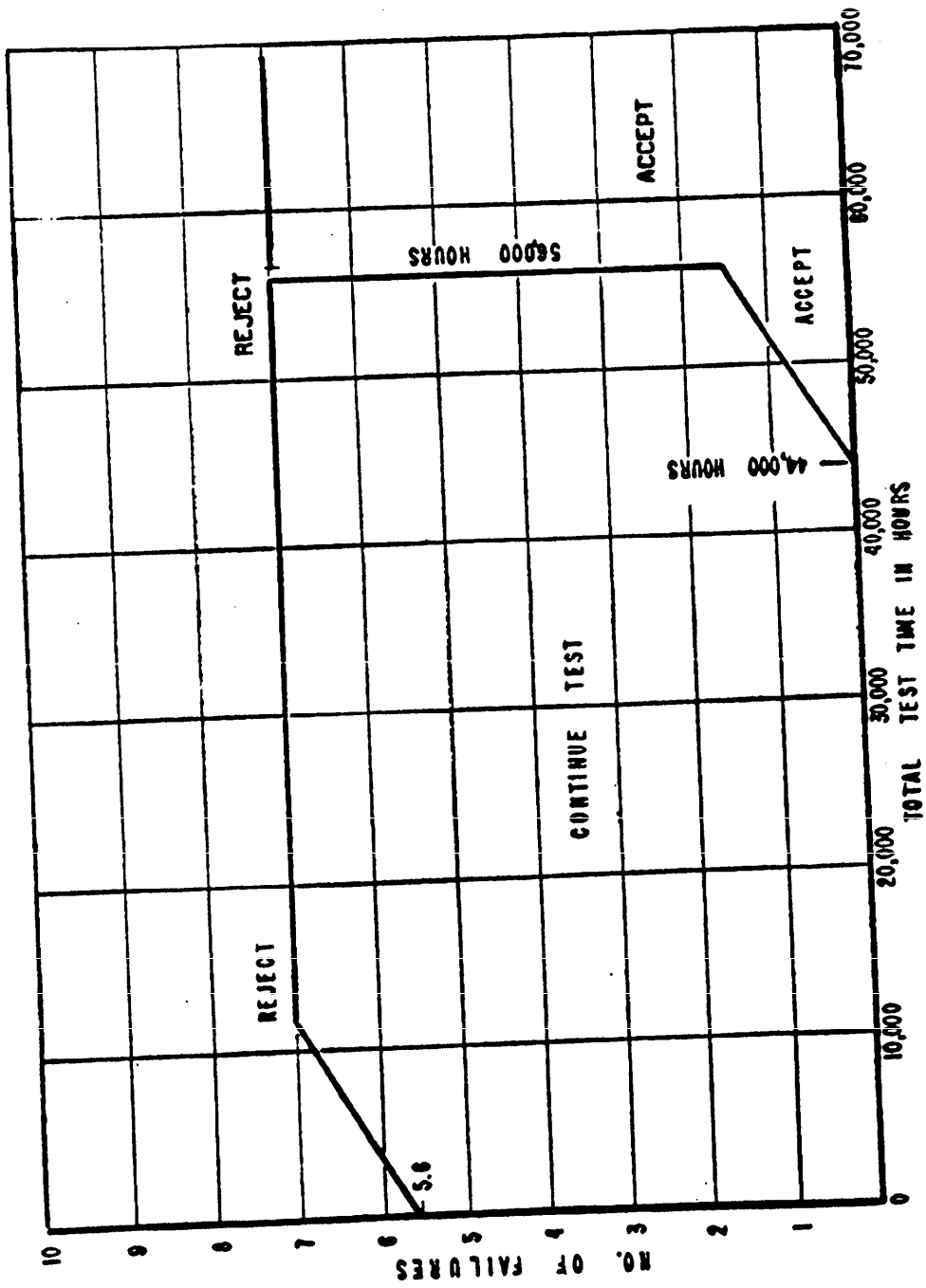


Figure 8. Controller Accept-Continue Test-Reject Criteria

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(b) Connect jumpers between the following connector pins:

7 to 5

7 to 6

(c) Excite rotors of all servoed repeaters from 400 cps phase A with R1 high and R2 low.

(d) Connect 400 cps input power through a normally open disconnect means to connector pins as follows:

Neutral to 21
115V phase A to 22
115V phase B to 23
115V phase C to 24.

4.7.2.1 Erection cycle and accuracy.- Align the displacement gyroscope in roll so that the two index surfaces on the clamp ring are parallel to a horizontal surface within 0.002 inch, and in pitch so that the longitudinal axis of the case is parallel to the horizontal surface within 0.002 inch.

(a) Apply power. After 1 minute, subject the displacement gyroscope to scorsby motion. Allow 2-1/2 minutes for completion of erection, note the pitch and roll attitude excursions at the end of this interval, and compute the average pitch and roll indications. Average pitch and roll indications shall be 0° within ±1/4°.

(b) Turn off scorsby motion. Tilt the displacement gyroscope 30° ±1° nose-up, followed by approximately 30° right-wing-down. Reapply scorsby motion.

(1) Record the average pitch and roll indications for one minute. Each shall be positive and approximately 30°.

(c) Without disturbing the displacement gyroscope attitude, remove the supply power for 20 seconds then reapply.

(1) The displacement gyroscope shall erect in a uniform fashion toward the average pitch and roll indications recorded in b (1).

(2) At 2-1/2 minutes after re-application of power, the average pitch and roll indications for one minute shall agree within ±1/4° with those recorded in b (1).

(d) Turn off scorsby motion and return the displacement gyroscope to level attitude.

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4.7.2.2 Heading Rotation. Rotate the case about the vertical axis in the direction of a right turn. The heading indicator shall show an increasing angle.

4.7.2.3 Heading drift rate.- Subject the displacement gyroscope to scorsby motion and record initial heading and the heading at the end of each of 3 consecutive 1/2-hour intervals. Compute the rate of heading change in degrees per hour for each interval and adjust these results to obtain free drift rate by adding the earth's rate correction for the latitude of test. The earth's rate correction is ±15 sine latitude degrees per hour (Northern Hemisphere). The results shall be as follows:

a. The average of the three hourly drift rates shall not exceed ±1/2° per hour.

b. The spread between the most positive and the most negative hourly rates shall not exceed 1° per hour.

4.7.2.4 Vertical drift rate.- Subject the displacement gyroscope to scorsby motion and test as follows:

a. Disconnect pitch erection by opening the jumper between pins 7 and 5 and simultaneously record pitch attitude. Record pitch attitude at the end of a 15-minute interval and compute rate of pitch change in degrees per hour. Adjust the result to obtain free drift rate by adding the earth's rate correction. The earth's rate correction in pitch is ±15 cosine latitude sine H degrees per hour, where H is the heading of the displacement gyroscope fore-aft axis with respect to geographic North. (H=0° for North, 90° for East.) The pitch drift rate shall not exceed 12° per hour. Reconnect the jumper between pins 7 and 5.

b. Disconnect roll erection by opening the jumper between pins 7 and 6 and simultaneously record roll attitude. Record roll attitude at the end of a 15-minute interval and compute rate of roll change in degrees per hour. Adjust the result to obtain free drift rate by adding the earth's rate correction. The earth's rate correction in roll is -15 cosine latitude cosine H degrees per hour. The roll drift rate shall not exceed 12° per hour. Reconnect the jumper between pins 7 and 6.

4.7.2.5 Erection rate.- During the erection rate measurement interval, the displacement gyroscope shall be subjected to scrosby motion.

a. Disconnect the roll axis servoed repeater and apply an isolated, center-tapped, 11.8V, phase A voltage to pins 13, 14, and 15 with the high side on 15 and the center tape on 14.

b. Disconnect pitch erection by opening the jumper between pins 7 and 5.

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c. Reverse pitch erection polarity by connecting 26V negative phase C to pin 5. Remove this negative phase C voltage when the pitch attitude becomes $\pm 10^{\circ}$ $\pm 1^{\circ}$ and remove the isolated 11.8 V phase A voltage from pins 13, 14, and 15.

d. Reapply pitch erection by reconnecting the 7 to 5 jumper, and begin timing when the pitch attitude becomes $\pm 8^{\circ}$. Note the pitch attitude at the end of 1 minute and compute the pitch erection rate by correcting for earth's rotation per 4.5.2.4a. The pitch erection rate shall be $5^{\circ} \pm 1^{\circ}$ per minute.

e. Reconnect the roll axis servoed repeater.

f. Disconnect the pitch axis servoed repeater, and apply an isolated center tapped 11.8V phase A voltage to pins 10, 11, and 12 with the high side on 12 and the center tap on 11.

g. Disconnect roll erection by opening the jumper between pins 7 and 6.

h. Reverse roll erection polarity by connecting 26V negative phase C to pin 6. Remove this negative phase C voltage when the roll attitude becomes $\pm 10^{\circ}$ $\pm 1^{\circ}$ and remove the isolated 11.8 V phase A voltage from pins 10, 11, and 12.

i. Reapply roll erection by reconnecting the 7 to 6 jumper, and begin timing when the roll attitude becomes $\pm 8^{\circ}$. Note the roll attitude at the end of 1 minute, and compute the roll erection rate by correcting for the earth's rotation per 4.5.2.4b. The roll erection rate shall be $5^{\circ} \pm 1^{\circ}$ per minute.

j. Reconnect the pitch axis servoed repeater.

4.7.2.5.1 Fast erection.- Repeat the procedure of 4.5.2.5 except:

a. In Steps c and h, tilt to attitude angles of $\pm 32^{\circ}$ $\pm 1^{\circ}$.

b. In Step d, apply pitch erection by connecting an 8 to 5 jumper. Replace by a 7 to 5 jumper upon completion of this step.

c. In Step i, apply roll erection by connectin an 8 to 6 jumper. Replace by a 7 to 6 jumper upon completion of this step.

d. In Steps d and i, begin timing at attitude angles of $\pm 30^{\circ}$. The erection rate shall be at least 12°/minute.

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4.7.2.6 Heading precession rate.- Subject the displacement gyroscope to secrsby motion.

a. Apply +1.100 ma ±0.005 ma dc with positive on North latitude control pin 17 and negative on ground pin 21. Determine the rate of heading change during a 30-minute interval, and correct the result for the earth's rotation to obtain heading precession rate.

(1) The heading precession rate shall be +9 ±3/4° per hour.

b. Disconnect the d-c input from pin 17.

c. Apply +11.0 ±0.1 ma dc with positive on South latitude control pin 18 and negative on ground pin 21. Determine the rate of heading change during a 6-minute interval and correct the result for the earth's rotation to obtain heading precession rate.

(1) The heading precession rate shall be -90 ±20° per hour.

d. Disconnect the d-c input from pin 18.

4.7.2.7 Linear acceleration sensor threshold.- Connect 26 ±0.25V 400 cps to pin 16 through a 6,000 ±1 percent ohm resistor. Measure the voltage from pin 16 to pin 21 using an a-c VTVM, Hewlett-Packard Model 400D, or equivalent.

a. Rotate the gimbal system in positive pitch by following the sequence defined in 4.7.2.5a, b and c, except remove the test voltages per 4.7.2.5c when the pitch angle has become approximately 5°.

b. Reapply pitch erection and monitor the pitch angle and the voltage indicated by the VTVM.

(1) the voltage shall increase to 12V at a positive pitch angle of 4°15' or less. Record the angle.

(2) The voltage shall increase to 15V at a positive pitch angle of 3°15' or greater. Record the angle.

(3) The voltage shall increase to 21V minimum when the gyroscope is fully erected in pitch. Record the voltage.

NOTE: Erection may be applied intermittently to facilitate testing.

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c. Fully erect the displacement gyroscope in pitch.

d. Rotate the gimbal system in negative pitch by following the sequence defined in 4.7.2.5a, b, and c, except reverse the polarity of the isolated 11.8V phase A connection in 4.7.2.5a and remove the test voltages per 4.7.2.5c when the pitch angle has become approximately 5°.

e. Reapply pitch erection and monitor the pitch angle and the voltage indicated by the VTVM.

(1) The voltage shall increase to 12V at a negative pitch angle of 4°15' or less. Record the angle.

(2) The voltage shall increase to 15V at a negative pitch angle of 3°15' or greater. Record the angle.

NOTE: Erection may be applied intermittently to facilitate testing.

f. Fully erect the displacement gyroscope in pitch.

4.7.2.8 Turn acceleration sensor threshold.- Connect 26 ±0.25V 400 cps to pin 9 through a 6,000 ±1 percent ohm resistor. Measure the voltage from pin 9 to pin 21 using an a-c VTVM, Hewlett-Packard Model 400D, or equivalent.

a. Rotate the gimbal system in positive roll by following the sequence defined in 4.7.2.5f, g, and h, except remove the test voltages per 4.7.2.5h when the roll angle has become approximately 10°.

b. Reapply roll erection and monitor the roll angle and the voltage indicated by the VTVM.

(1) The voltage shall increase to 12V at a positive roll angle of 4°15' or less. Record the angle.

(2) The voltage shall increase to 15V at a positive roll angle of 3°15' or greater. Record the angle.

(3) The voltage shall increase to 21V minimum when the displacement gyroscope is fully erected in roll. Record the voltage.

NOTE: Erection may be applied intermittently to facilitate testing.

c. Fully erect the displacement gyroscope in roll.

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d. Rotate the gimbal system in negative roll by following the sequence defined in 4.7.2.5f, g, and h, except reverse the polarity of the isolated 11.8V phase A connection in 4.7.2.5c and remove the test voltages per 4.7.2.5h when the roll angle has become approximately 10°.

e. Reapply roll erection and monitor the roll angle and the voltage indicated by the VTVM.

(1) The voltage shall increase to 12V at a negative roll angle of 4°15' or less. Record the angle.

(2) The voltage shall increase to 15V at a negative pitch angle of 3°15' or greater. Record the angle.

NOTE: Erection may be applied intermittently to facilitate testing.

f. Fully erect the displacement gyroscope in roll.

4.7.2.9 Fast Erection Warning Signal.-

a. Disconnect power to displacement gyroscope.

b. After 1 minute , apply power to the displacement gyroscope.

c. Record voltage on pin 4 one minute after power application. Voltage should be 78V±3V.

d. Record voltage on pin 4 2½ minutes after power application. Voltage should be 26V±2V.

4.7.2.10 Pitch gimbal freedom.-

a. Rotate displacement gyroscope 90° in positive pitch and record the maximum value measured by the pitch servoed repeater. The maximum value should be +83° ±2°.

b. Level displacement gyroscope.

c. Rotate displacement gyroscope 90° in negative pitch and record the maximum value measured by the pitch servoed repeater. The maximum value should be 83° ±2°.

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4.7.2.11 Torsional response.- Subject the gyroscope, including the vibrator isolators to sinusoidal angular oscillations of $2^{\circ} \pm .2^{\circ}$ peak to peak. about the roll pitch and heading axes independently. The response characteristics at 5, 10. and 15 cps shall be as follows: using performance at .5cps to represent unity amplitude and 0 degrees phase lag.

CPS/sec	Max Amp	Max Phase Lag
5	1.13	5°
10	1.38	15°
15	2.50	60°

This test is part of Sampling Plan "B".

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4.7.3 Controller tests

4.7.3.1 Mode selector. The mode selector shall be tested as follows:

Measure the resistance between the specified connector pins. The resistance shall be within the following limits:

Switch Position	Pins	Resistance
DG --	5 and 23	1 megohm min
DG	24 and 23	1 megohm min
Slaved	5 and 23	1 ohm max
Slaved	24 and 23	1 megohm min
Compass	5 and 23	1 megohm min
Compass	24 and 23	1 ohm max

4.7.3.2 Hemisphere selector. The hemisphere selector shall be tested as follows:

a. Measure the resistance between connector pins 9 and 10. The resistance shall be within the following limits:

Switch Position	Resistance
N	20K ohm max
S	1 megohm min

b. Measure the resistance between connector pins 8 and 10. The resistance shall be within the following limits:

Switch Position	Resistance
N	1 megohm min
S	20K ohm max

4.7.3.3 Latitude control current. Connect one 1,000 ohm ±0.1 percent resistor from pin 8 to pin 10 and another from pin 9 to pin 10. Apply +27 ±1V dc with negative on pin 10 and positive in series with an 820 ±0.2 percent ohm resistor to pin 4.

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a. Place hemisphere select switch in N position and connect an adjustable resistance unit having a 0.1 percent accuracy from +27V to pin 8. Set latitude control and adjustable resistance to the following values and read d-c voltage from pin 8 to pin 9. The voltage must not exceed ±30 millivolts.

Latitude Angle (Degrees)	Resistance (Ohms)
00.0	open
05.0	167,750
10.0	83,906
15.0	55,962
20.0	42,062
25.0	33,839
30.0	28,444
35.0	24,665
40.0	21,920
45.0	19,833
50.0	18,231
60.0	16,003
70.0	14,670
80.0	13,958
90.0	13,730

Transfer the hemisphere select switch to the S position and transfer the pin-8 end of the adjustable resistance unit to pin 9. Set latitude control and adjustable resistance to values tabulated and read d-c voltage from pin 8 to pin 9. Voltage must not exceed ±30 millivolts.

Latitude Angle (Degrees)	Resistance (Ohms)
00.0	open
30.0	28,444
90.0	13,730

Disconnect all test equipment.

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4.7.3.4 Synchronization indicator deflection

a. Apply +27 ±1V dc through a 56.2K ohm resistor with positive on pin 6 and negative on pin 7.

(1) The sync indicator must deflect halfway toward + .

b. Repeat step a, except with polarity reversed. The sync indicator will deflect halfway toward - .

c. Disconnect all test equipment.

(1) The sync indicator must return to center ±1/16 inch.

4.7.3.5 Heading set operation. Connect 26 ±1V 400 cps from pin 3 to pin 23.

a. Measure resistance from pin 1 to pin 23. Resistance shall be within the following limits for each position of the heading set knob:

Knob Position	Resistance
Depressed	1 ohm max
Released	1 megohm min

b. Repeat step a, measuring resistance from pin 12 to pin 23.

c. Measure a-c voltage from pin 2 to pin 23. Voltage shall be within the following limits for each position of the heading set control:

Knob Position	Voltage
Depressed and full cw	4 to 5V
Depressed and full ccw	4 to 5V
Released	75 mv max

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d. Measure a-c voltage from pin 3 to pin 2 with control depressed and rotated fully ccw

(1) Voltage must be 30.5 ±2V

e. Reduce applied voltage to 2.5 ±0.2V

(1) Output voltage from pin 2 to pin 23 shall be less than 100 mv when knob is depressed and rotated full cw and full ccw.

f. Disconnect all test equipment.

4.7.3.6 Malfunction indicator

a. Apply 26V 400 cycles from pin 11 to pin 23

(1) Malfunction indicator must be lighted

b. Disconnect the 26V 400 cycles from pin 11 to pin 23 and reconnect from pin 3 to pin 23

c. Push the malfunction indicator

(1) The malfunction indicator must be lighted and must remain lighted as long as the indicator is depressed

d. Remove the 26V 400-cycle voltage.

4.7.3.7 Panel lighting. (This test may be conducted prior to final assembly.)

a. Apply 5 ±0.1V ac/dc on pins 21 and 22

(1) Panel lighting must comply with MIL-P-7788

b. Remove all test connections.

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4.7.4 Electronic control amplifier tests. Connect the simulated controller to the amplifier using the connections of Table IV. Connect the simulated displacement gyroscope heading transmitter stator leads to the following connector pins per table III:

S1 to 1
S2 to 2
S3 to 3

Connect the simulated azimuth detector stator leads to the following connector pins per table IV:

S1 to 16
S2 to 17
S3 to 18

Connect the servoed repeater to each heading output as required using the following connector pins per table V. Output No. 1 is to be monitored unless specifically stated otherwise.

Output No. 1	Output No. 2
S1 to 1	S1 to 4
S2 to 2	S2 to 5
S3 to 3	S3 to 6
R1 to 13	R1 to 15
R2 to 14	R2 to 16

Output No. 3	Output No. 4
S1 to 7	S1 to 10
S2 to 8	S2 to 11
S3 to 9	S3 to 12
R1 to 17	R1 to 19
R2 to 18	R2 to 20

Connect the simulated gyroscope transmitter stator leads to the following connector pins per table III:

For Pitch Channel Tests

S1 to 19
S2 to 20
S3 to 21

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For Roll Channel Tests

S1 to 22
S2 to 23
S3 to 24

Connect a 1000 ±0.1 percent resistor between each of the following connector pins per table III:

From 17 to 9

From 18 to 9

Connect 400-cps power to the following connector pins per table V.

Neutral to 23

115V ϕA to 24

115V ϕB to 25

115V ϕC to 26

4.7.4.1 Slaving amplifier output and gain. Set the controls of the controller and amplifier as follows:

Mode switch to SLAVED

Latitude control to 0°

Hemisphere switch to N

Compensation switch to OFF

a. Set simulated azimuth detector to 0° and the displacement gyroscope heading transmitter to 0°.

b. Synchronize the system by depressing and holding the heading set knob until the servoed repeater stops rotating. This synchronization should take a maximum of six seconds.

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c. Note the deflection of the synchronization indicator, and measure the d-c voltage with positive on pin 18 and negative on pin 17 of connector J2 of table III.

(1) The sync indicator must be centered with ±1/16 inch.

(2) The d-c voltage from pin 18 to pin 17 must be 0 ±1V.

d. Position the gyroscope transmitter to the following headings. The sync indicator must deflect toward POSITIVE to the proportion of scale noted, and the d-c voltage must agree with the tabulation

Gyroscope Heading	Deflection	Volts
0° 15'	Part Scale	---
3°	---	+11 ±2
4°	Full scale	---
90°	Full scale	+11 ±3

e. Position the azimuth detector to 90° and re-synchronize the system, leaving the displacement gyroscope transmitter at 90°.

(1) The sync indicator must be centered within ±1/16 inch.

(2) The d-c voltage from pin 18 to pin 17 must be 0 ±1V.

f. Position the azimuth detector to the following headings. The sync indicator must deflect toward NEGATIVE to the proportion of scale noted, and the d-c voltage must agree with the tabulation.

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Simulated Azimuth Detector Heading	Deflection	Volts
90° 15'	Part scale	---
93°	---	-11 ±2
94°	Full scale	---
180°	Full scale	-11 ±3

g. Switch to DG mode

(1) The sync indicator must become centered with ±1/16 inch.

(2) The d-c voltage from pin 18 to pin 17 must become 0 ±1V.

h. Set the latitude control to 90.0°

(1) The sync indicator must remain centered with ±1/16 inch.

(2) The d-c voltage from pin 18 to pin 17 must become -1.833 ±0.050V.

4.7.4.2 Heading compensation range. Set the controls as follows:

Mode switch to SLAVED

Latitude control to 0°

Hemisphere switch to N

Compensation switch to OFF

a. Set all compensation potentiometers approximately at center

b. Turn compensation switch on.

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c. Rotate 0° compensation potentiometer fully cw. Synchronize the system and position the azimuth detector to obtain a 0° heading.

d. Turn compensation switch off and re-synchronize the system.

(1) The heading must have decreased by 4° ±1° from the compensated value.

e. Turn compensation switch on.

f. Rotate 0° compensation potentiometer fully ccw. Synchronize the system and position the azimuth detector to obtain a 0° heading.

g. Turn compensation switch off and re-synchronize the system.

(1) The heading must have increased by 4° ±1° from the compensated value.

h. Return the 0° compensation potentiometer approximately to center.

i. Repeat steps b, c, d, e, f, g, h, and i with the heading set at each 15° and with the correspondingly marked compensation potentiometer being employed.

4.7.4.3 Heading servo scale error DG mode. Set the controls as follows:

Mode switch to DG

Latitude control to 0°

Hemisphere switch to N

Compensation switch to OFF

a. Set the gyroscope heading transmitter to 0°.

b. Using the heading set drive, position the servoed repeater connected to heading output No. 1 to 0°.

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c. Check the heading indicated by output No's 2, 3, and 4.

(1) Each indicated heading shall be $0^{\circ} \pm 6'$.

d. Position the gyroscope transmitter to each 30° beginning at 0° and record the indicated heading for output No. 1.

e. Determine the point errors by subtracting the gyroscope heading from the indicated heading at each position (negative numbers may occur).

f. Calculate the scale error spread by determining the maximum algebraic difference between the point errors.

(1) The scale error spread shall not exceed 30 minutes.

g. Repeat steps d, e, and f for heading output No's 2, 3, and 4.

(1) The scale error spread for output No's 2 and 3 shall not exceed 30 minutes.

(2) The scale error spread for output No. 4 shall not exceed 36 minutes.

4.7.4.4 Heading servo scale error, compensated, slaved mode. Set the controls as follows:

Mode switch to SLAVED

Latitude control to 0°

Hemisphere switch to N

Compensation switch to ON

a. Set all compensation potentiometers approximately at center.

b. Set the azimuth detector heading to 0° 23' in accordance with the first row of the following tabulation.

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c. Set the indicated heading of output No. 1 to 0° as tabulated, using the gyroscope heading transmitter.

d. Adjust the 0° compensation potentiometer to produce a null voltage of 0 ±0.5V dc from pin 17 to pin 18 of connector J2.

e. Repeat steps b, c, and d for each of the following tabulated heading pairs, in sequence:

Azimuth Detector Heading	Indicated Heading
0° 23'	0°
16° 09'	15°
31° 49'	30°
47° 23'	45°
62° 46'	60°
77° 58'	75°
92° 58'	90°
107° 46'	105°
122° 23'	120°
136° 49'	135°
151° 09'	150°
165° 23'	165°
179° 37'	180°
193° 51'	195°
208° 11'	210°
222° 37'	225°
237° 14'	240°
252° 02'	255°
267° 02'	270°
282° 14'	285°
297° 37'	300°
313° 11'	315°
328° 51'	330°
344° 37'	345°

f. A second compensation is permitted beginning with the compensation potentiometers positioned as in e (this would consist of repeating steps b, c, d, and e).

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g. Re-set the azimuth detector to each heading as tabulated, and set the indicated heading to produce a null voltage of 0 ±0.5 V dc from pin 17 to pin 18 of connector J2.

(1) The indicated heading must agree with the tabulation within ±9 minutes.

h. Re-center the compensation potentiometers and turn the compensation switch off.

4.7.4.5 Heading servo follow-up rate. Set the controls as follows:

Mode switch to DG

Latitude control to 0°

Hemisphere switch to N

Compensation switch to OFF

a. Rotate the gyroscope heading transmitter at a rate of 300° per second minimum for 1 minute.

(1) The indicated heading shall rotate smoothly.

4.7.4.6 Heading set rate. Set the controls as follows:

Mode switch to SLAVED

Latitude control to 0°

Hemisphere switch to N

Compensation switch to OFF

a. Set the azimuth detector to 0° ±6'.

b. Push and hold the heading set control until the indicated heading becomes synchronized with the azimuth detector heading. Record the indicated heading. It should read 0° ±30'.

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c. Hold the heading set control depressed but not deflected and rotate the gyroscope heading transmitter in an increasing heading direction at a rate of 30° per second.

(1) The indicated heading must increase by no more than 5° from the value recorded in step b.

d. Discontinue rotation of the gyroscope heading transmitter and release the heading set control.

e. Switch to DG mode.

f. Push the heading set control and rotate it cw to contact the stop.

(1) The indicated heading must increase at 30° per second minimum.

g. Measure resistance from pins 13 to 15 and 14 to 15 connector J3 with the heading set control depressed and with it released.

(1) Resistance shall be within the following limits:

Heading Set Control Position	Resistance 13 to 15	Resistance 14 to 15
Depressed	1 ohm max	1 megohm min
Released	1 megohm min	1 ohm max.

4.7.4.7 Malfunction monitoring circuit performance. Set the controls as follows:

Mode switch to DG

Latitude control to 0°

Hemisphere switch to N

Compensation switch to OFF

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a. Increase the supply voltage (keeping all phases balanced) until a malfunction is signalled. (Note: do not exceed 138V line-to-neutral)

(1) A malfunction must be signalled as specified in 3.6.6.1 for a supply voltage in the range of 126.5 to 138V line-to-neutral.

b. Return the supply voltage to 115 ±1V line-to-neutral.

(1) The malfunction signals must disappear.

c. Decrease the supply voltage (keeping all phases balanced) until a malfunction is signalled.

(1) A malfunction must be signalled as specified in 3.6.6.1 for a supply voltage in the range of 103.5 to 92V line-to-neutral.

d. Return the supply voltage to 115 ±1V line-to-neutral.

(1) The malfunction signals must disappear.

e. Position the gyroscope heading transmitter to 0°.

f. Set the indicated heading to 0°.

g. Remove the fixed field excitation from the heading servo motor.

h. Slowly rotate the gyroscope heading transmitter in a positive direction until a malfunction is signalled as specified in 3.6.6.1.

(1) A malfunction must be signalled within the region of +2° to +3°.

i. Slowly return the gyroscope heading transmitter to 0°.

(1) The malfunction signals must have disappeared when the gyroscope heading becomes +2°.

j. Reconnect the heading servo motor fixed phase excitation.

k. Remove the excitation from the gyroscope heading transmitter.

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(1) A malfunction must be signalled as specified in 3.6.6.1.

l. Reconnect the gyroscope heading transmitter excitation.

(1) The malfunction signals must disappear.

m. Disconnect the gyroscope heading transmitter stator leads in pairs from connector J2 pins 1, 2, and 3.

(1) A malfunction must be signalled as specified in 3.6.6.1.

(2) The malfunction signals must disappear with all leads connected.

4.7.4.8 Compass mode operation. Set the control as follows:

Mode switch to COMP

Latitude control to 0°

Hemisphere switch to N

Compensation switch to OFF.

Rotate the azimuth detector in 30° increments noting the heading output. The difference between the azimuth detector position and the heading output shall not vary by more than 1° at each point.

4.7.4.9 Linear acceleration control functions. Set the controls as follows:

Mode switch to SLAVED

Latitude control to 0°

Hemisphere switch to N

Compensation switch to OFF

a. Set a decade resistor to 8,000 ohms and connect it from pin 5 to pin 9 of connector J2.

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b. Position the azimuth detector to 0°.

c. Push and hold the heading set control until the indicated heading becomes synchronized with the azimuth detector heading.

d. Re-position the azimuth detector to 10°.

e. Note the sync indicator deflection.

(1) The sync indicator must be at negative full scale.

f. Measure a-c voltages at the tabulated pins of connector J2.

(1) Voltages shall be within the following limits:

Connector Pins	Volts
7 to 9	26 ± 2
7 to 12	89 ± 3

g. Set the decade resistor to 5,000 ohms.

(1) The a-c voltage from pin 7 to pin 9 must become 6.5 ±1V.

(2) The sync indicator must become centered within ±1/16 inch.

h. Set the decade resistor to 8,000 ohms.

(1) The a-c voltage from pin 7 to pin 9 must return to 26 ±2V.

(2) The sync indicator must return to negative full scale.

4.7.4.10 Turn acceleration cutoff function. Set the controls as follows:

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Mode switch to SLAVED.

Latitude control to 0°.

Hemisphere switch to N.

Compensation switch to OFF.

a. Set a decade resistor to 8,000 ohms and connect it from pin 6 to pin 9 of connector J2.

b. Position the azimuth detector to 0°.

c. Push and hold the heading set control until the indicated heading becomes synchronized with the azimuth detector heading.

d. Re-position the azimuth detector to 10°.

e. Note the sync indicator deflection.

(1) The sync indicator must be at negative full scale.

f. Measure a-c voltages at the tabulated pins of connector J2.

(1) Voltages shall be within the following limits:

Connector Pins	Volts
8 to 9	6.5 ±1
8 to 12	108.5 ±3

g. Set the decade resistor to 5,000 ohms.

The sync indicator must become centered within ±1/16 inch.

h. Set the decade resistor to 8,000 ohms.

The sync indicator must return to negative full scale.

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4.7.5. System operational tests

4.7.5.1 Test connections. Interconnect all system components in accordance with figure 9, using the simulated azimuth detector of 4.2.2.5. Connect a servoed repeater to the heading, roll, or pitch system output as required. Connect 400-cps power to the following pins of connector 2J1:

Neutral to 23
115V ϕA to 24
115V ϕB to 25
115V ϕC to 26

4.7.5.2 Special test condition. Wait a minimum of 2-1/2 minutes after application of power before conducting any test. Controls for the controller and the amplifier shall be set as follows except where otherwise noted:

Mode switch to DG.

Latitude control to local latitude.

Hemisphere switch to local hemisphere.

Compensation switch to OFF

4.7.5.3 Test procedures. Unreasonable results shall be investigated by performing the applicable individual test on the component involved. If heading drift trim adjustment is performed, the heading drift rate test for the gyroscope shall be repeated.

4.7.5.3.1 Heading rotational polarity

4.7.5.3.1.1 DG mode

a. Rotate the gyroscope through a right turn heading change of $90^\circ \pm 6'$.

(1) The transmitted heading signal from the amplifier shall increase by $90^\circ \pm 1.5^\circ$.

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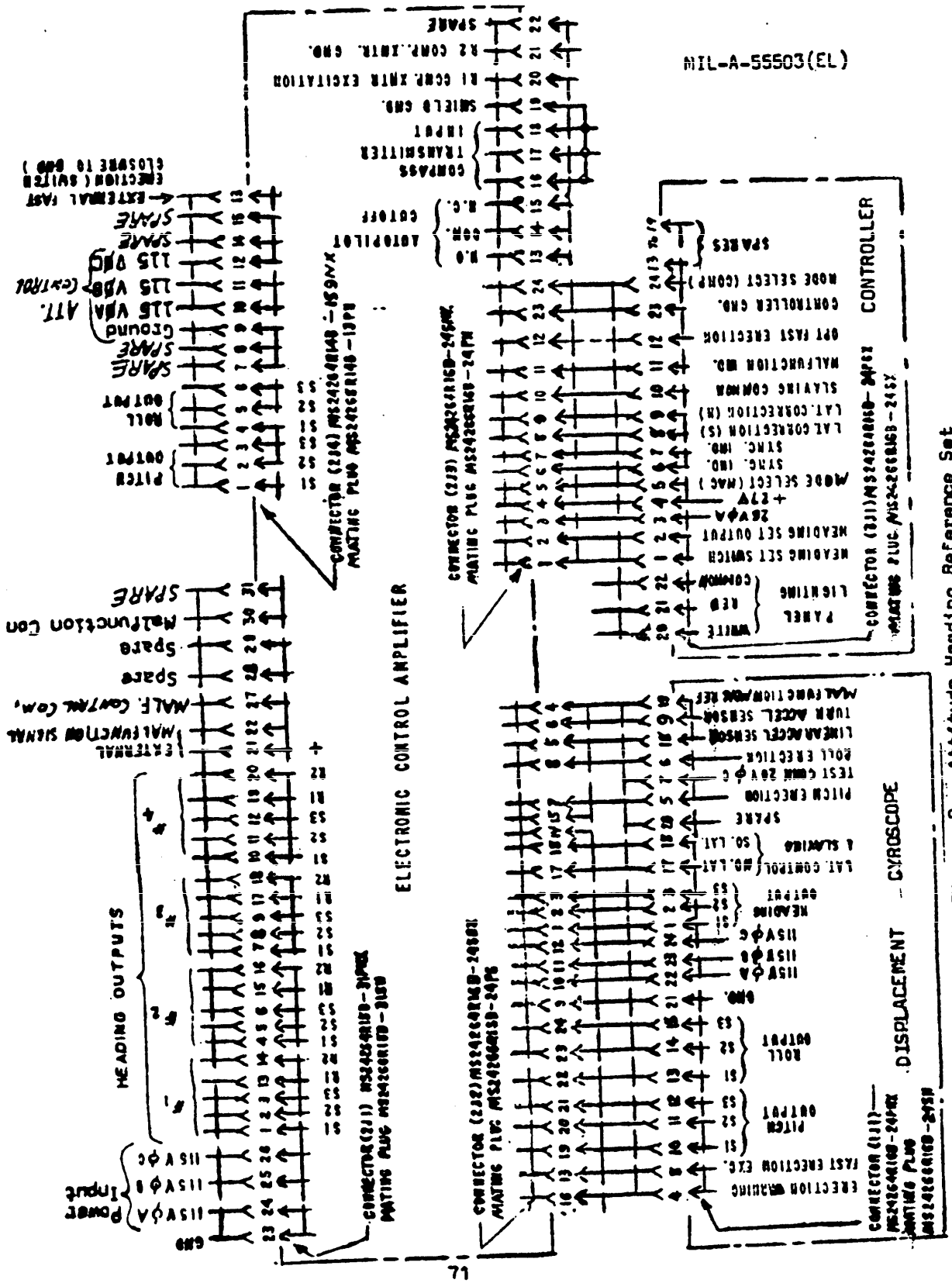


Figure 9 Attitude Heading Reference Set

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4.7.5.3.1.2 Slaved mode

a. Transfer the mode switch to SLAVED.

b. Synchronize the system by depressing and holding the heading set control until the servoed repeater stops rotating. Synchronization should occur within 6 seconds.

(1) Note that the sync indicator is centered ±1/16 inch.

c. Rotate the displacement gyroscope through a right-turn heading change of 60° ±6'.

d. Rotate the azimuth detector through a right-turn heading change to produce a synchronization indicator null.

(1) Null must be achieved with a right-turn heading rotation of 60° ±1.5°.

e. Hold the heading set control depressed and rotate the azimuth detector through a right-turn heading change of 90° ±6'.

(1) The transmitted heading signal from the amplifier shall increase to 90° ±1°.

4.7.5.3.1.3 Compass mode

a. Transfer the mode switch to COMP.

(1) The sync indicator shall become centered and the system heading shall stabilize.

b. Rotate the displacement gyroscope through a turn of 60° ±10°.

(1) The system heading shall not change.

c. Rotate the azimuth detector through right and left turns of 60° ±2°.

(1) The system heading output shall follow the azimuth detector smoothly to 60° ±2°.

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d. Return the mode switch to DG.

4.7.5.3.2 Roll correspondence and rotational polarity

a. Position the gyroscope to a right-wing-down roll angle of 30° ±6'.

(1) The transmitted roll signal from the amplifier shall be +30° ±1°.

b. Return the gyroscope to a level position.

4.7.5.3.3 Pitch correspondence and rotational polarity

a. Position the gyroscope to a nose-up pitch angle of 30° ±6'.

(1) The transmitted pitch signal from the amplifier shall be +30° ±1°.

b. Return the gyroscope to a level position.

4.7.5.3.4 Heading set function

4.7.5.3.4.1 DG mode. Depress the heading set control and deflect it cw. The transmitted heading signal must increase at a rate which is adjustable, with low rate corresponding to small deflection angle.

4.7.5.3.4.2 Slaved mode

a. Transfer the mode switch to SLAVED.

b. Synchronize the system by depressing and holding the heading set control.

(1) Once the transmitted heading signal has stopped rotating, it shall remain stable while the control remains depressed.

c. Transfer mode switch to DG.

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d. Depress the control, deflect it cw for 1 to 3 seconds, then release it without returning it to center.

e. Transfer mode switch to SLAVED.

(1) The sync indicator shall show a positive deflection.

f. Return the mode switch to DG.

4.7.5.3.5 Slaving rate and correspondence

a. Transfer the mode switch to SLAVED.

b. Synchronize the system.

c. Rotate the gyroscope through a 5° right turn.

(1) The sync indicator shall show full scale positive deflection.

d. Rotate the azimuth detector through a 10° ±6' right turn.

(1) The sync indicator shall show full scale negative deflection.

e. Re-synchronize the system and note the indicated heading 10° ±1°.

f. Rotate the gyroscope to change the heading 15° from the synchronized null position, and allow normal slaving to return the heading toward the null position.

(1) The time required for the heading to change from the 15° to the 5° off null position shall be between 5 and 10 minutes.

g. Return the mode switch to DG.

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4.7.5.3.6 System heading drift rate. Subject the gyroscope to scorsby motion and set the latitude control to the local latitude.

a. Record the system heading; then record it again at the end of a 1-hour interval.

(1) The change in heading shall be $\pm 1^{o}$ maximum..

4.7.6 Voltage and frequency variation. For this test and all remaining tests, inter connection of system components, test equipment and power, and settings of system controls shall be in accordance with 4.7.2 through 4.7.6, as applicable. The components shall be properly connected and operated at normal voltage and frequency, and shall be set to a synchronized null with the simulated azimuth detector at 0° heading. The voltage and frequency extremes shall be applied and the system shall be re-synchronized at each value. The shift in the synchronized null shall not exceed $3/4^{o}$. The system or components, as applicable, shall meet the requirements of 4.7.2.3 with a 10 percent increase in tolerance, of 4.7.2.5 with a 20% increase in tolerance, of 4.7.5.3.6 with a 50% increase in tolerance, and of 4.7.4.5, with an input rate of 45 rpm. No damage shall result from the voltage and frequency extremes.

4.7.7 Power consumption. The equipment shall give specified performance from a category B power source as defined in MIL-STD-704. The power required at nominal voltage and frequency shall not exceed the following values, exclusive of power supplied from this equipment to operate associated external equipment.

a. A-C power (three phase) 115/200V, 400 cps.

(1) Displacement gyroscope

75w - starting
50w - running

(2) Control amplifier and the controller

50w

b. AC/DC power 5V

(1) Controller lighting

6.6w

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4.7.8 Vibration error. The system shall be operated in the slaved mode, synchronized to the simulated azimuth detector, and subjected to a vibration error test. During this test, the input vibration to the equipment shall coincide with a power spectral density of $0.015g^2/cps$ for each frequency from 20 to 2,000 cps. In order to accomplish this representation, the loading effect of the unit under test shall be eliminated by proper load compensation. This random vibration shall be applied to the equipment along the three mutually perpendicular axes of the component. The signal source shall be either a prepared magnetic tape or a noise generator. To properly observe errors, the duration shall be from 10 to 15 minutes. The system outputs of pitch, roll, and heading shall not vary by more than 3/4° from the initial position established with 10 percent of this vibration level, and shall return to the initial position when the vibration is again reduced to 10 percent of the test vibration level.

4.7.8.1 DG mode. The system shall be operated in the DG mode and subjected to the vibration error test. During this test, the input vibration to the equipment shall coincide with a power spectral density of $0.002g^2/cps$ for each frequency from 20 to 2,000 cps. In order to accomplish this representation, the loading effect of the units under test shall be eliminated by proper load compensation. This random vibration shall be applied to the equipment along the three mutually perpendicular axes of each component. The signal source shall be either a prepared magnetic tape or a noise generator. The duration of the test shall be 15 minutes. The system heading change shall not exceed $\pm 1/2^\circ$ when corrected for the earth's rotation.

4.7.9 Low field strength. The system shall be connected to a simulated azimuth detector and operated on a synchronized null. The excitation of the simulated azimuth detector shall be reduced to 102 ±2.5 mv 800 cps to represent a horizontal field strength of 0.080 ±0.002 oersted. The simulated azimuth detector shall be oscillated through ±5° either side of null at 6 cpm. The system heading shall be noted each minute after start of the oscillation until three consecutive readings are within 0.1° of each other. The difference between the initial system heading and the final system heading shall not exceed 0.75°. The test shall be run on three headings not less than 60° apart.

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4.7.10 Slaved dynamic accuracy. The system shall be operated in the slaved mode and synchronised to the simulated azimuth detector. The simulated azimuth detector shall be oscillated through ±5° either side of the null at 6 cpm, and readings of the heading output shall be made at 1-minute intervals. Readings shall continue until three consecutive 1-minute readings are within 0.1° of each other, which shall occur within 45 minutes from the start of the oscillation. The difference between the initial and final headings shall not exceed 1.5°.

4.7.11 Radio noise interference. Radio noise interference testing shall be in accordance with MIL-I-6181. The system shall be properly connected with the gyroscope operated on a 15° scorsby mechanism during the test and with the lighting circuit energised.

4.7.12 Temperature-altitude (cycling). The system components shall be subjected to the temperature-altitude test specified in MIL-T-5422 for class 2 equipment. Maximum test temperature with equipment operating shall be 71°C. Maximum nonoperating temperature for the controller shall be 85°C. Tests to determine satisfactory operation shall consist of 4.7.2.1, step a, and 4.7.5.3.5 with tolerances increased as noted. Tests to determine performance characteristics shall consist of 4.7.2.1, step a, 4.7.2.4, and 4.7.5.3.6 with tolerances increased as noted. At the conclusion the units shall successfully complete the individual tests with no tolerance increases.

a. 4.7.2.1, step a - 5 minutes will be allowed for completion of erection when at reduced temperatures

b. 4.7.2.4 - Limits are ±15° per hour.

c. 4.7.5.3.6 - Limit is ±2° per hour.

4.7.13 Humidity. The components shall be subjected to the humidity test specified in MIL-T-5422. Upon completion, the gyroscope shall meet the requirements of 4.7.2.1 and the other components shall meet the individual tests. There shall be no evidence of detrimental corrosion or rusting of external parts of the system components.

4.7.14 Fungus. The components shall be subjected to the fungus test specified in MIL-T-5422. Upon completion of this test, there shall be no evidence of detrimental deterioration, corrosion, or fungus growth on any external surface of the components.

4.7.15 Rain. The controller and amplifier shall be subjected to a rain test in accordance with procedure II of MIL-T-5272. At the conclusion of this test, the controller and amplifier shall meet the tests specified in 4.7.4.5. There shall be no evidence of water damage.

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4.7.16 Salt spray. The components shall be subjected to the salt spray test specified in MIL-T-5422. Upon completion of this test, the components shall meet the tests specified in 4.7.2.1, step a, 4.7.4.5, and 4.7.5.3.5. There shall be no evidence of corrosion that would shorten the operational life of the components.

4.7.17 Sand and dust. The controller and amplifier shall be subjected to the sand and dust test specified in MIL-T-5422. Upon completion of this test, the controller and amplifier shall meet the tests specified in 4.7.4.5. There shall be no evidence of damage due to sand and dust.

4.7.18 Temperature shock. The components shall be subjected to the temperature shock test specified in MIL-E-5272. Upon completion of this test, the components shall meet the tests specified in 4.7.2.1, 4.7.4.5, and 4.7.5.3.5.

4.7.18.1 Leak rate. The gyroscope seal shall not permit the equivalent of more than a 10 percent loss of the filling medium after 1,000 hours at a pressure differential equal to a 100,000-foot altitude.

4.7.19 Vibration failure. The components shall be subjected to the following vibration tests:

4.7.19.1 Controller. The controller shall be subjected to the vibration test in MIL-T-5422 (Curve I is applicable) and during both the cycling and resonance vibration test specified in MIL-T-5422 the controller circuits shall be checked to assure proper continuity. At the end of this test, the controller shall meet the tests specified in 4.7.3.

4.7.19.2 Gyroscope. The gyroscope shall be subjected to the vibration test specified in MIL-T-5422 (Curve I is applicable) and during both the cycling and resonance vibration test specified in MIL-T-5422 the pitch and roll servos transmitters shall not exceed one (1) degree with normal erection applied and the heading drift shall not exceed one (1) degree during any half hour interval. At the completion of this test, the gyroscope shall meet the tests specified under 4.7.2.

4.7.19.3 Amplifier. The amplifier shall be subjected to the vibration test specified in MIL-T-5422 (Curve I is applicable) and during the cycling and resonance vibration test the slaving circuit shall be checked per 4.7.4.1 and the heading servo scale error checked per 4.7.4.3. When not performing the above test the simulated gyroscope transmitter shall be rotated uniformly at approximately 5 RPM and the output shall follow smoothly. At the end of this test, the amplifiers shall meet the tests specified under 4.7.4.

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4.7.20 Explosion test. The nonhermetically sealed components shall be subjected to an explosion test in accordance with procedure I of MIL-T-5422.

4.7.21 Shock. The components shall be subjected to a shock test in accordance with MIL-T-5422. Upon completion of this test, the components shall meet the individual tests.

4.7.22 Crash safety. The components shall be subjected to a crash safety test in accordance with MIL-T-5422.

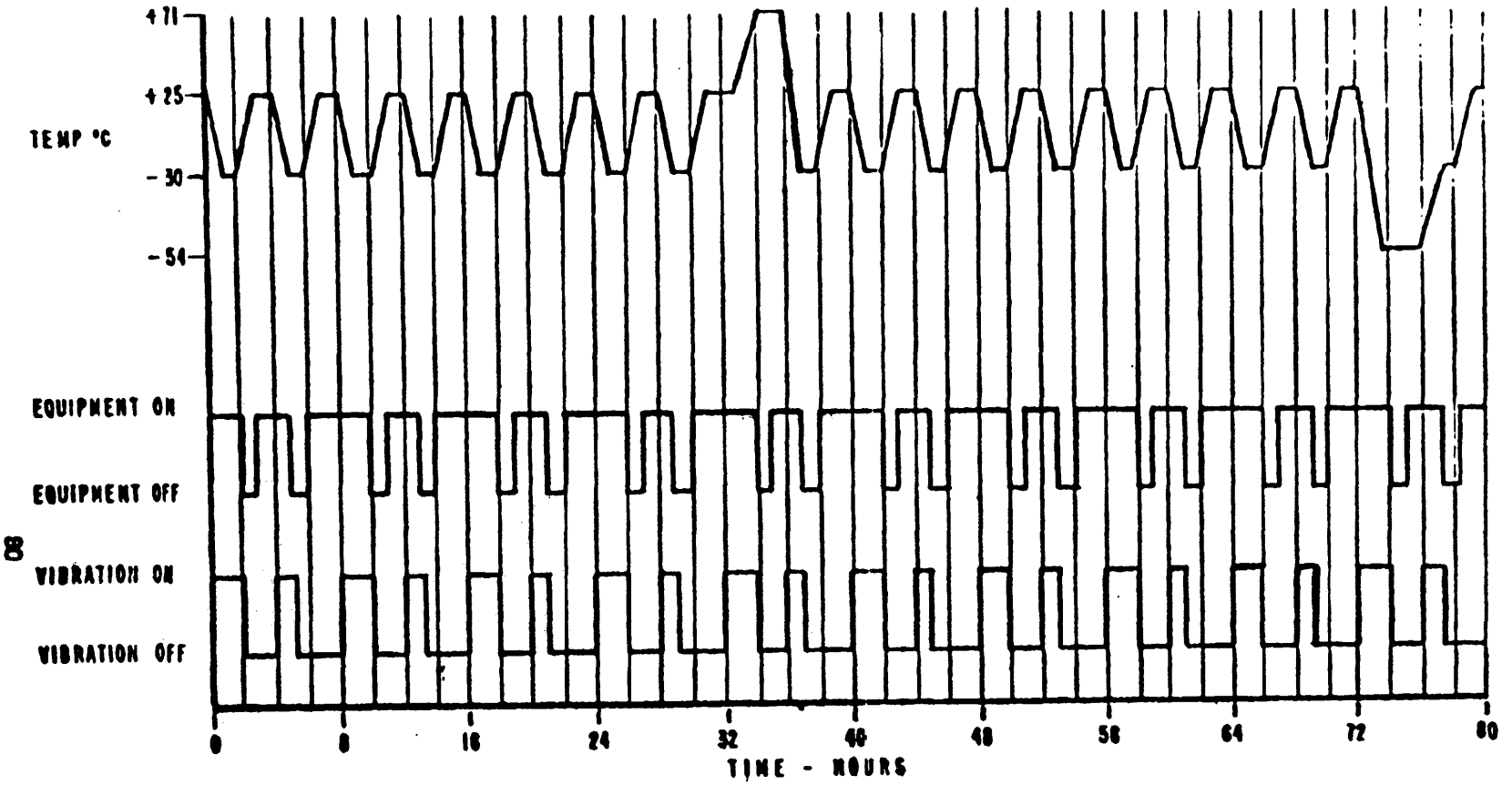
4.7.23 Reliability. The components shall be properly connected as shown on figure 9 and operated to assure proper setup. The Attitude Heading Reference Set or the components shall be placed in test chambers which meet the environmental cycle conditions shown on figure 10. During the ON cycle, the displacement gyros shall be exercised by a 3-axis, 15° total movement, 5 to 7 cpm, reversing scoreby action and by the vibration shown on figure 10. The other components shall be vibrated as shown on figure 10. After each 160 hour cycle of operation, the components shall meet the appropriate individual tests and the attitude heading reference set shall meet Sampling Plan A. The reliability test shall be conducted and test reports prepared and submitted in accordance with MIL-STD-781A, however, figures 5 through 8 shall be the accept or reject criterior for MTBF test and figure 10 shall be the applicable environmental cycle for this test.

4.7.23.1 Longevity. At the completion of the reliability test specified in 4.7.23, three Attitude Heading Reference Sets or six components from each production sample shall continue to be tested for a total of 2,000 hours each. The components shall meet the appropriate individual tests and the Attitude Heading Reference Set shall meet Sampling Plan A.

4.7.23.2 Allowances and failures. The increase in tolerances allowed for reliability and longevity testing shall be as follows:

Percent Increase in Tolerance	Reliability Test Time Per System or Component
0	0 to 500 hours
25	500 to 1,000 hours
50	1,000 hours and over

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1) Temperature change > 2°C/minute average. Periods of +71°C & -54°C, tests may be changed to coincide with working shift, with Government approval.
2) Equipment off period = 1/2 hour within the period shown. (Equipment shall be on at least ½hr. at temp. extremes.)
3) Vibration on period = 3 hrs/8 hrs approx as shown.
4) Temp. chamber shall be stabilized at the temp. shown at each hour point.
5) Environment cycle repeats at each 80 hours.
6) Vibration level = 0.2g 20 to 30 cps.
7) Monitoring of outputs for catastrophic failure each "on" cycle.
8) Monitoring of performance conducted each 160 hours.
9) Accumulated hours = all operating time.

Figure 10. Reliability Test Environment Cycles

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Out-of-tolerance calibration-type requirements will only be considered as failures if they occur in two successive cycles unless the equipment fails to meet the requirements by an additional 10%; then they will be considered failures with only one occurrence. Except for the above reservations, a failure shall be as defined in 6.3.2.

4.7.24 Heading Drift Rate Adjustment.- Determine the heading drift rate in accordance with 4.7.2.3 at both extremes of the drift adjustment. The difference between the drift rates at the two extremes shall be 4.6^{0} per hour $\pm 1^{0}$ per hour.

5 Preparation for Delivery

5.1 Preparation for delivery shall be in accordance with instructions of the procuring activity.

6 NOTES

6.1 Intended use.- The attitude-heading reference set covered by this specification is intended for use in aircraft to provide pitch, roll, and heading displacement signals of the aircraft for indication and tie-in to other equipment. Provisions are made for various types of operation and error compensation, depending upon the application and associated equipment.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Whether a mounting rack for the Electronic Control Amplifier will be provided

c. Whether sampling plan C tests will not be conducted

d. Whether sampling plan D test will not be conducted

e. Whether test reports are not required or when test reports are to be submitted.

f. All failures occurring during testing to be reported to the procuring activity within 2 working days

6.3 Definitions

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6.3.1 Hermetic seal.- A hermetic seal is defined as a perfectly closed and airtight seal made between vitric or metallic materials, or both. A hermetic seal is not intended to include seals accomplished by gaskets or O-rings.

6.3.2 Failure.- Where a condition is specified as a requirement, a failure will be considered as any electrical or mechanical condition which fails to meet the requirements specified herein.

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